



KubeCon



CloudNativeCon

Europe 2024

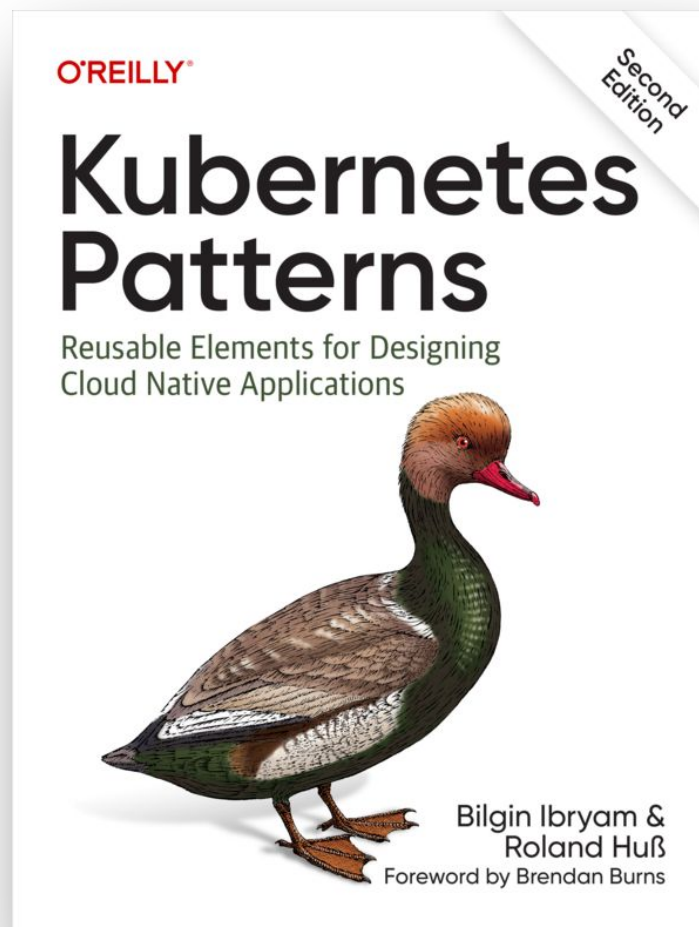


10 Years of Kubernetes Patterns Evolution

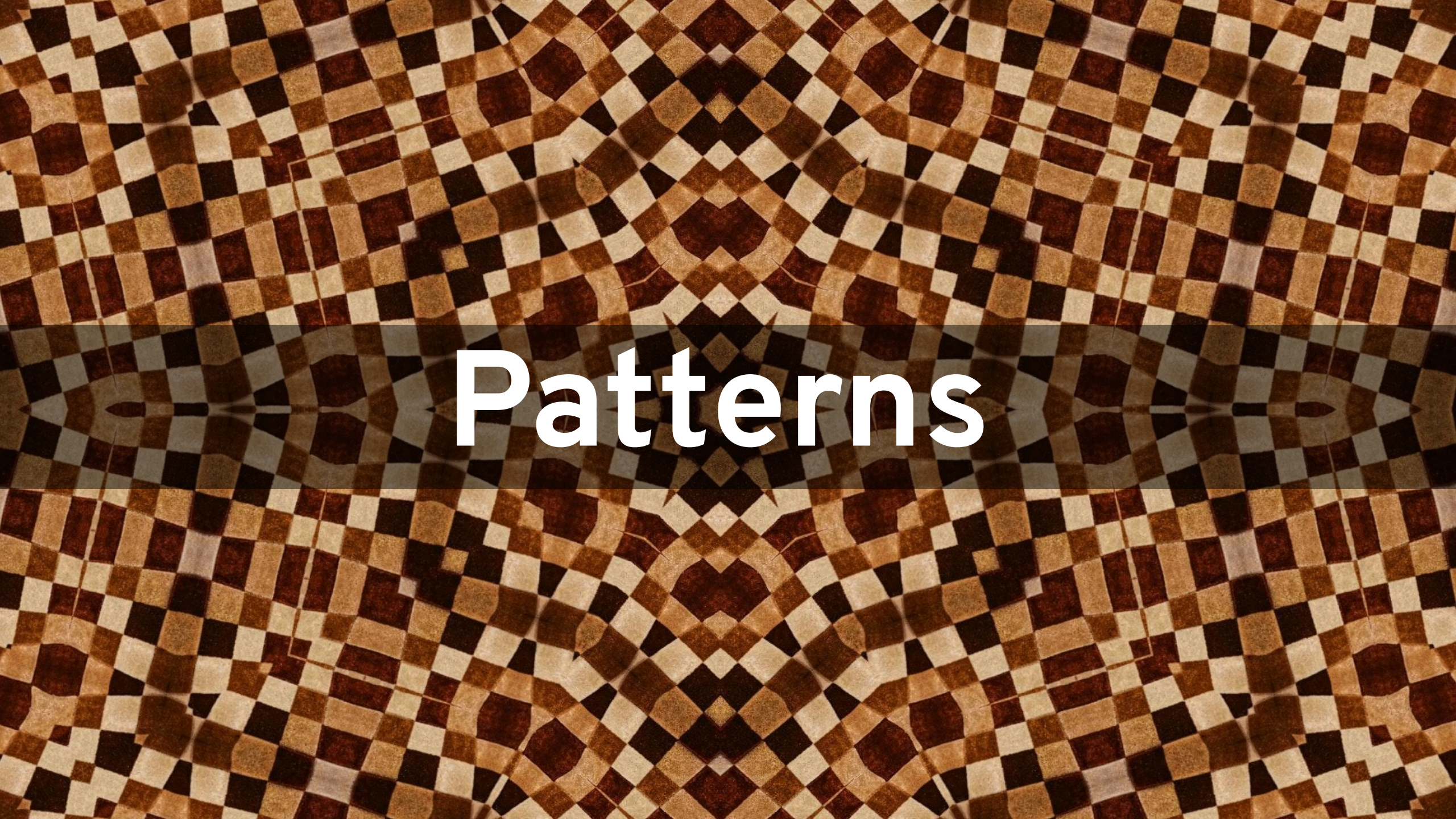
Bilgin Ibryam, Diagrid, @bibryam

*Roland Huß, Red Hat,
@rol4nd@hachyderm.io*

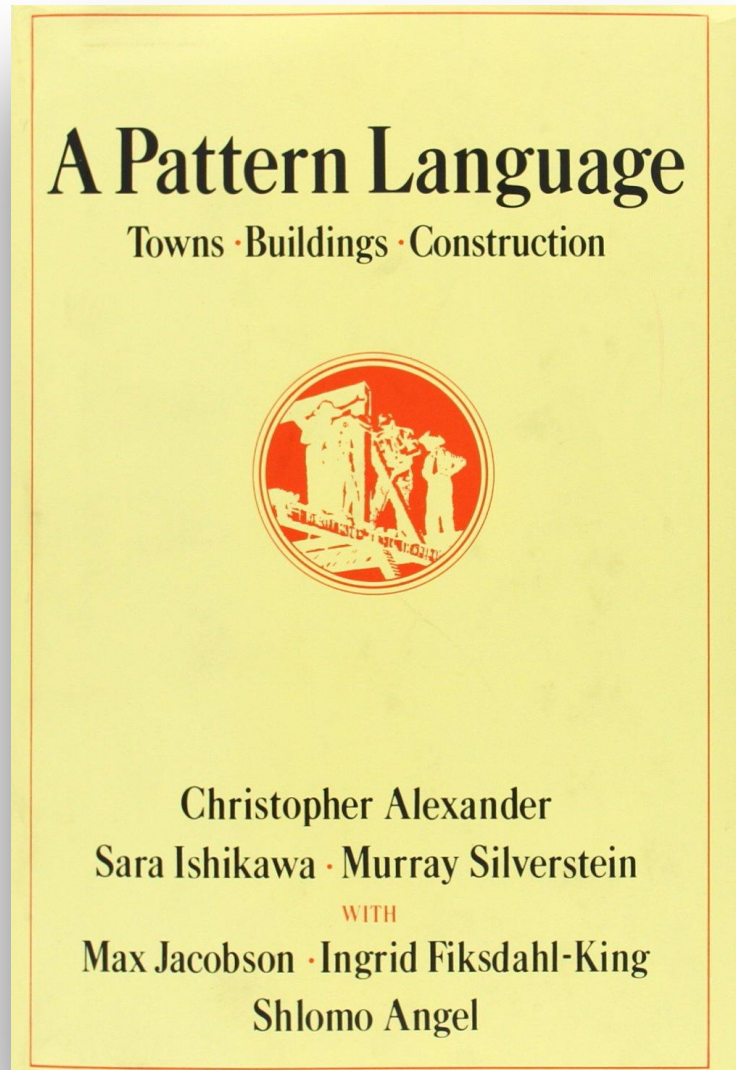
k8spatterns.com

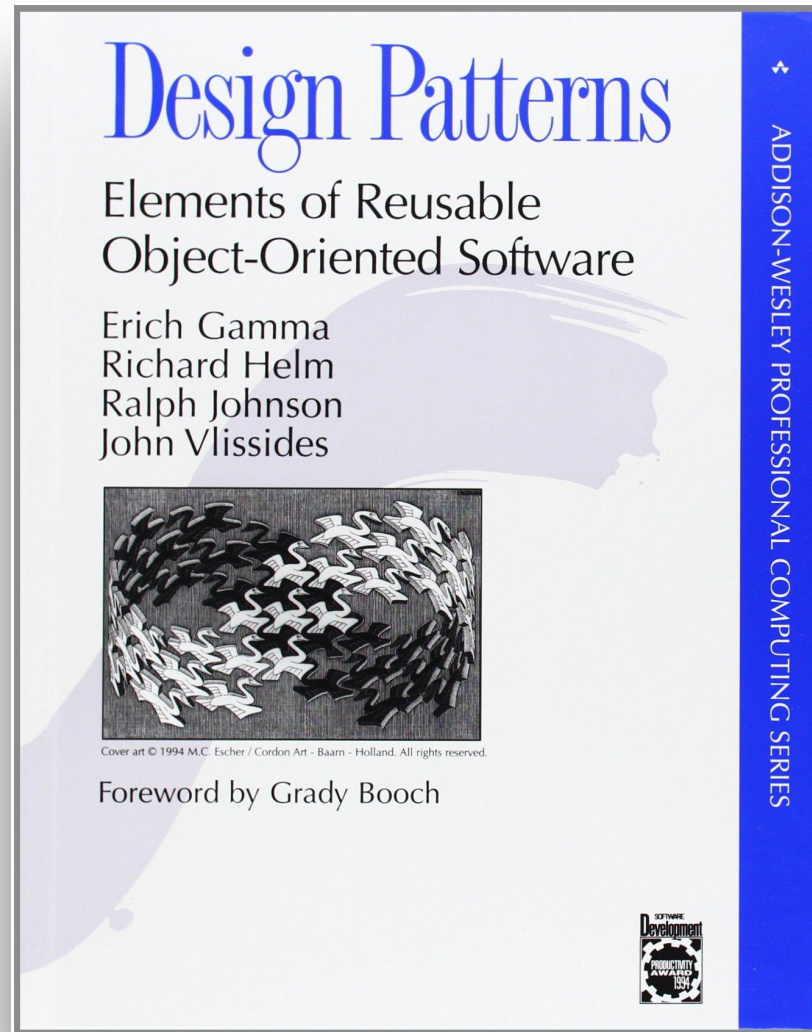


<https://k8spatterns.com>



Patterns



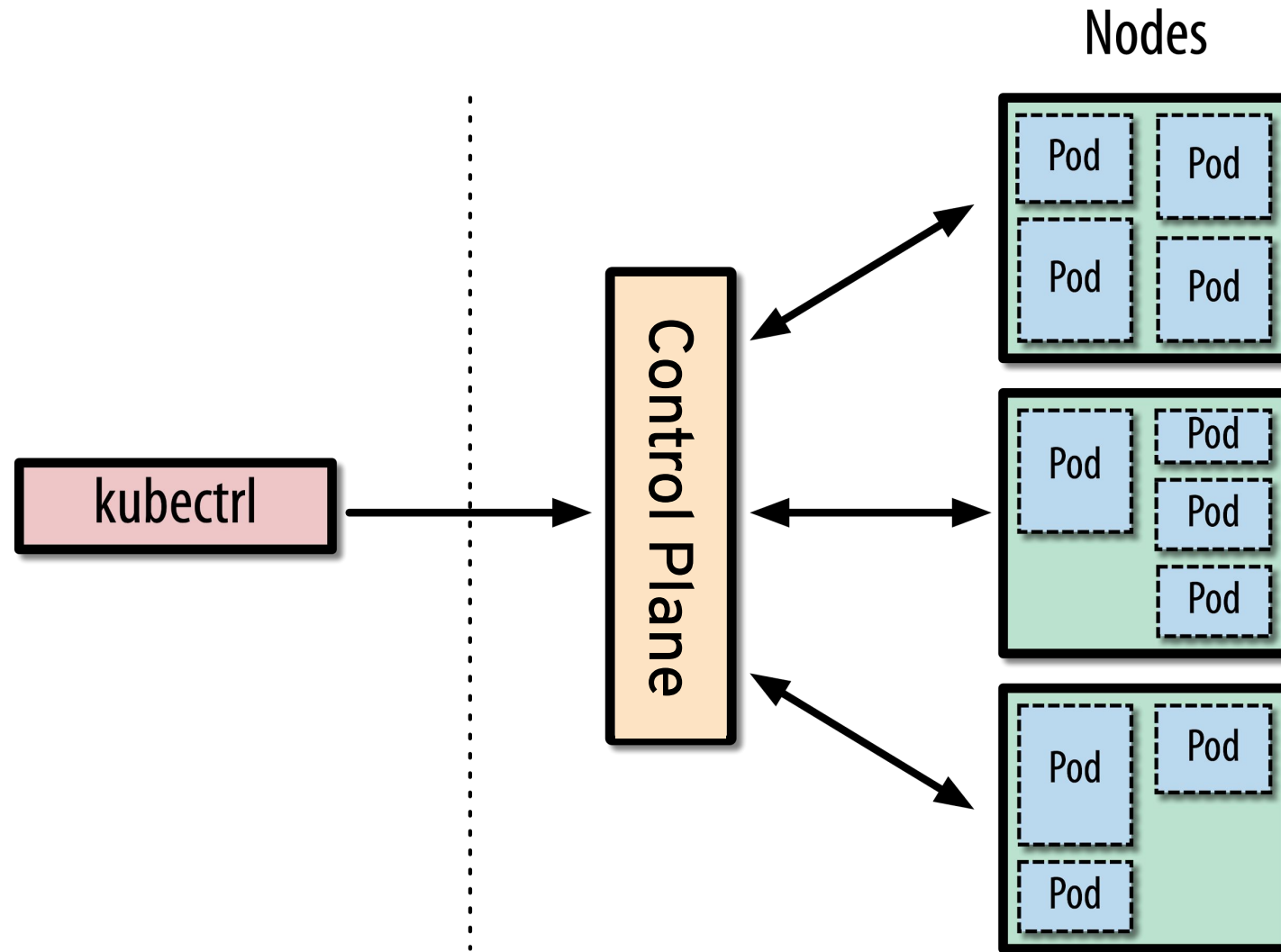


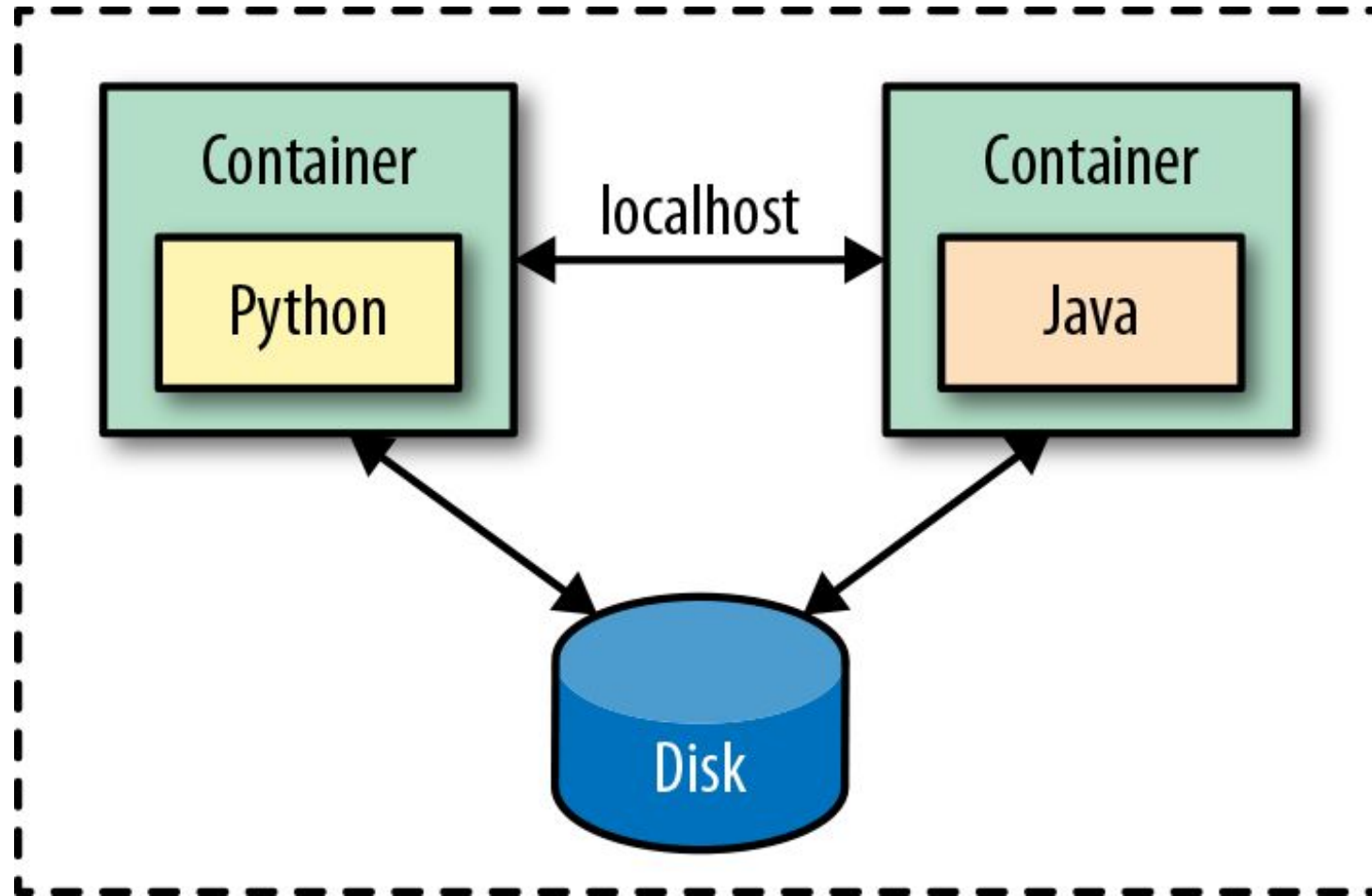


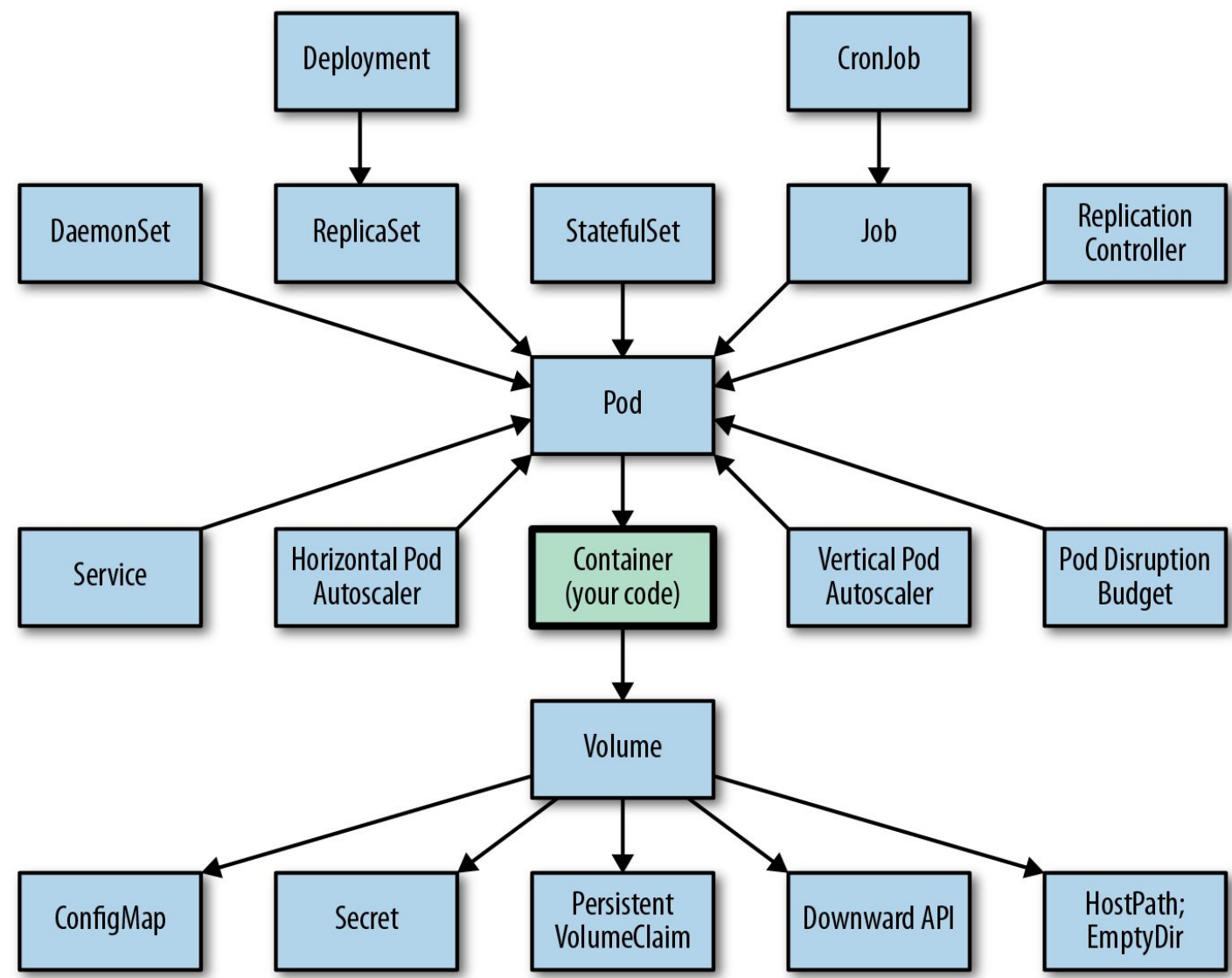
Kubernetes



- Open Source container orchestration system started by Google in 2014
 - ✧ Scheduling
 - ✧ Self-healing
 - ✧ Horizontal and vertical scaling
 - ✧ Service discovery
 - ✧ Application Deployments
- Declarative resource-centric REST API







Pattern Categories



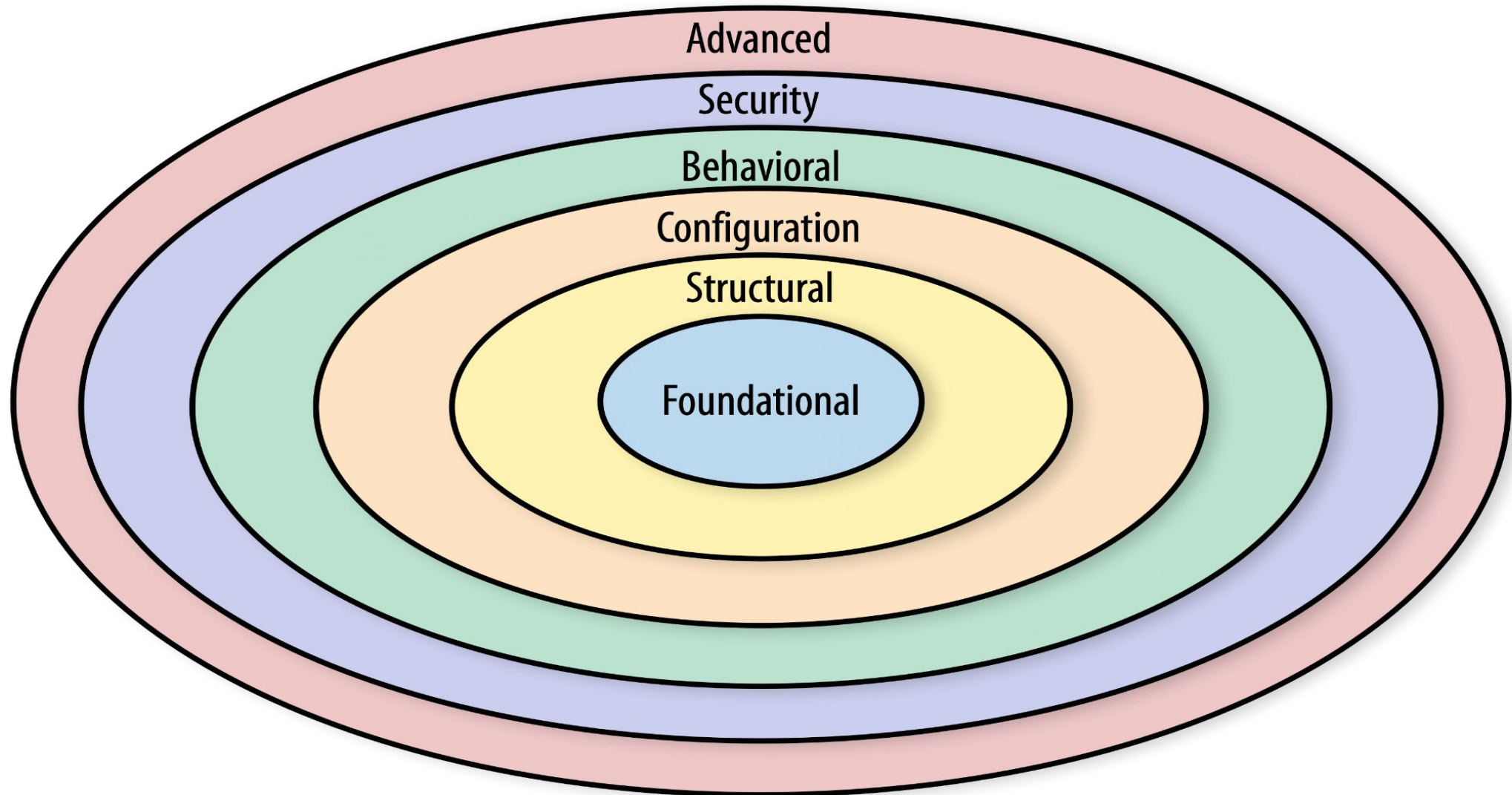
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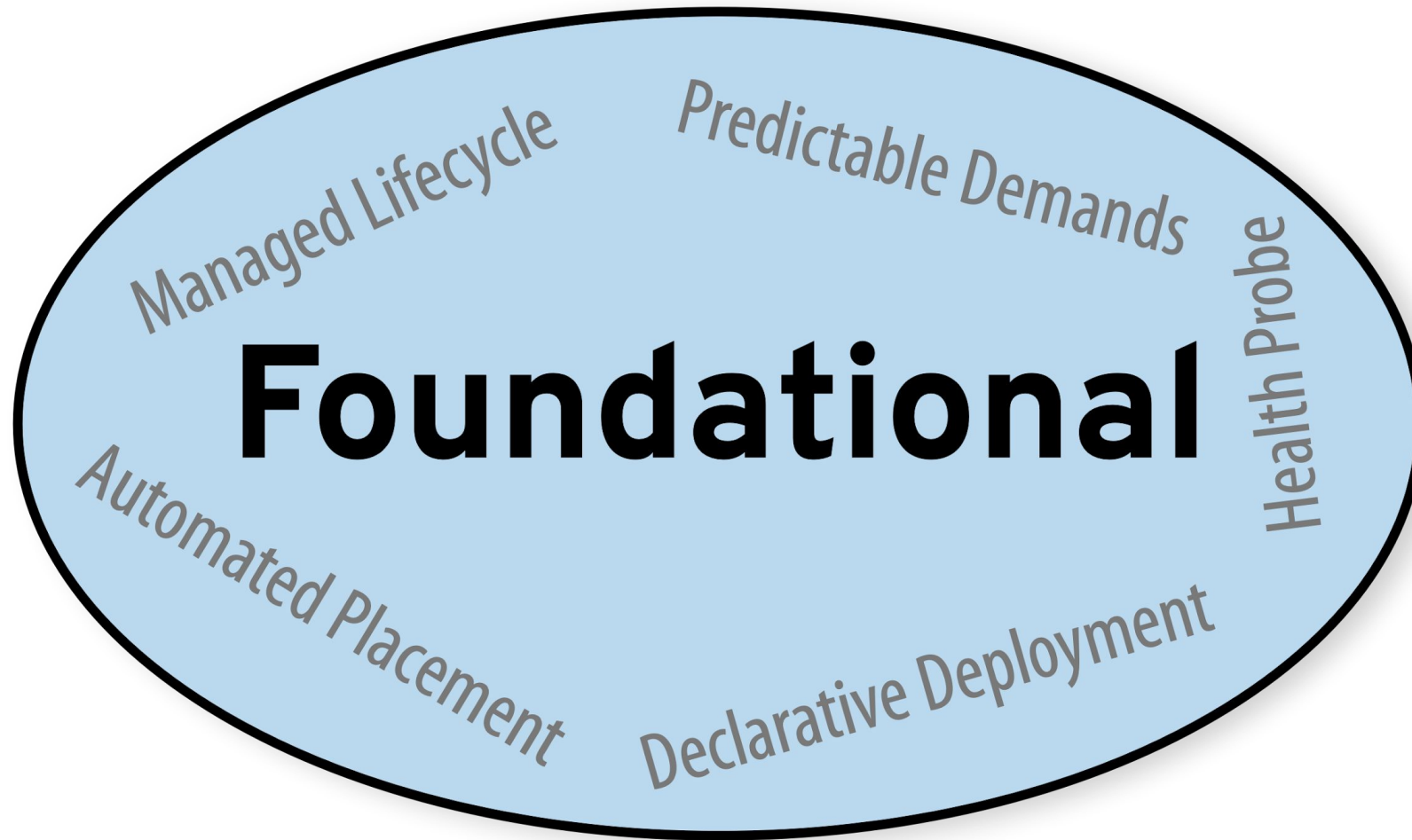


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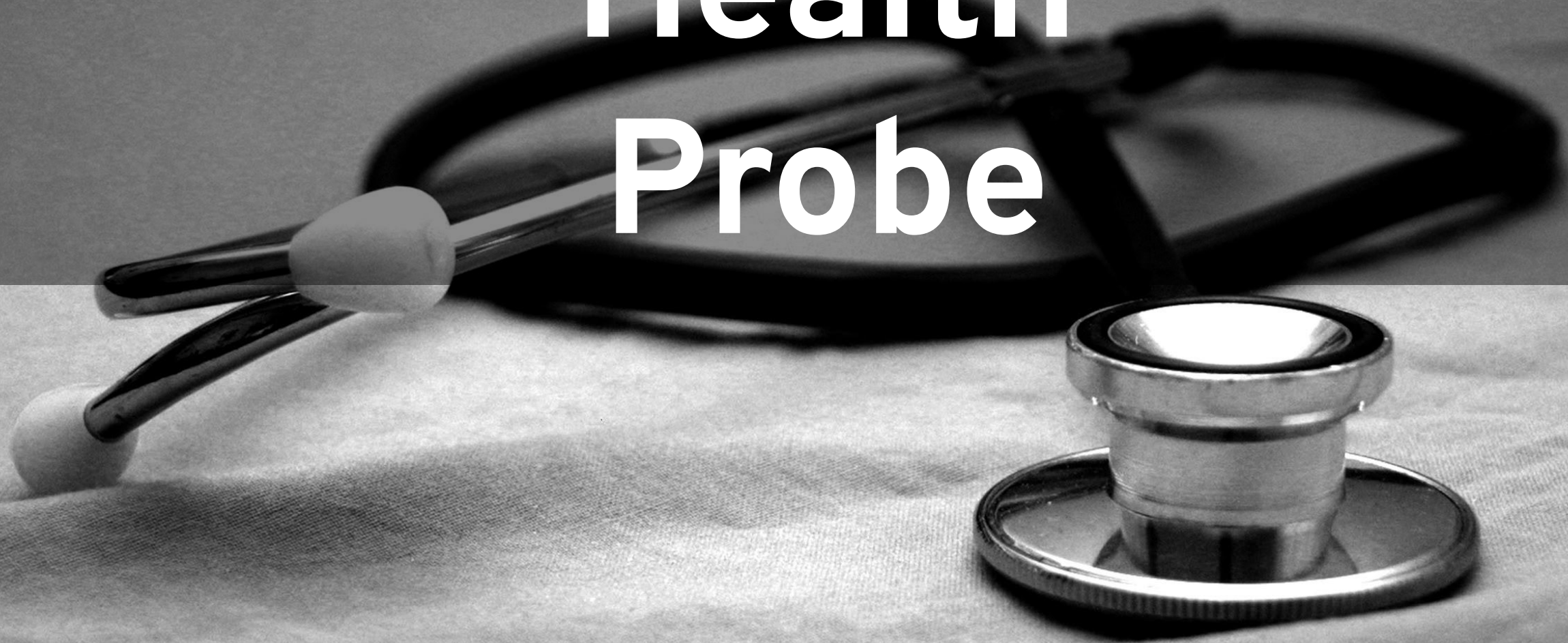
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Health Probe



How to communicate an application's health state to Kubernetes

Container Observability Options



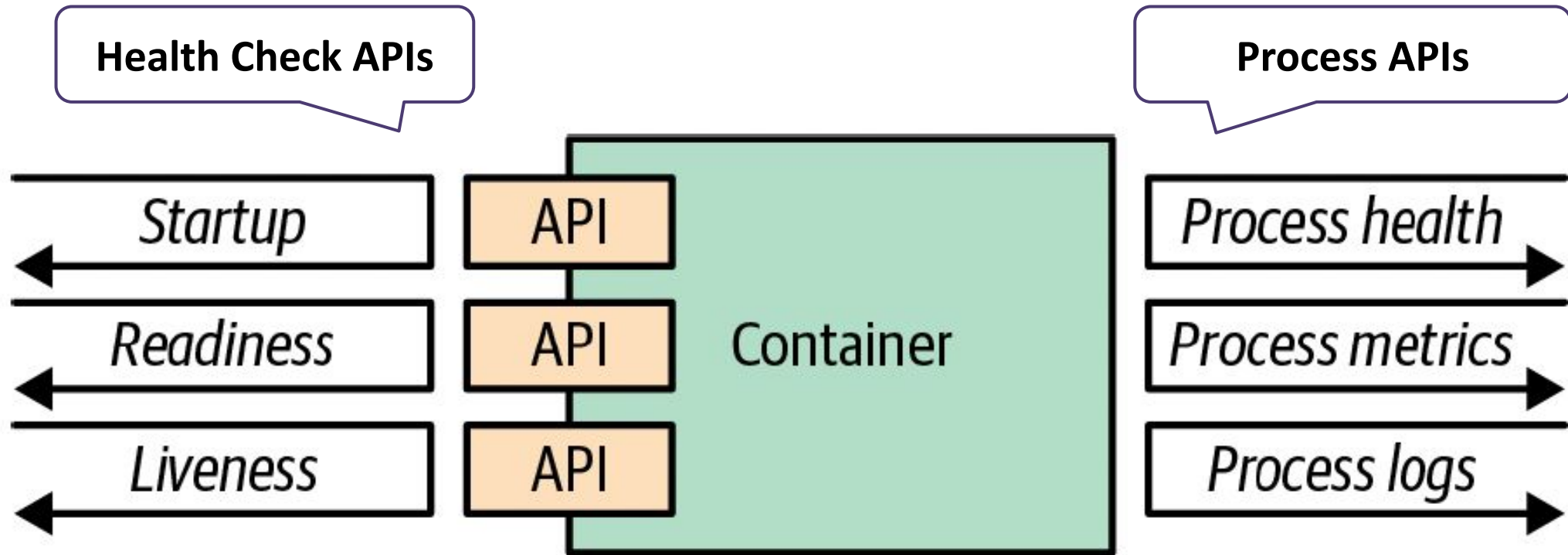
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- Liveness Probe
 - Restarting containers if liveness probes fails
- Readiness Probe
 - Removing from service endpoint if readiness probe fails
- Startup Probe
 - Restarting containers if startup probes fails
- Probe methods
 - HTTP endpoint
 - gRPC endpoint
 - TCP socket endpoint
 - Unix command return value

Health Probe Example



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```
containers:
- image: k8spatterns/random-generator:1.0
  name: random-generator
  livenessProbe:
    httpGet:
      path: /actuator/health
      port: 8080
  readinessProbe:
    grpc:
      port: 5000
  startupProbe:
    exec:
      command: [ "stat", "/opt/jboss/wildfly/tmp/startup-marker" ]
    initialDelaySeconds: 30
    periodSeconds: 60
    failureThreshold: 15
```

Foundational Patterns Dependencies



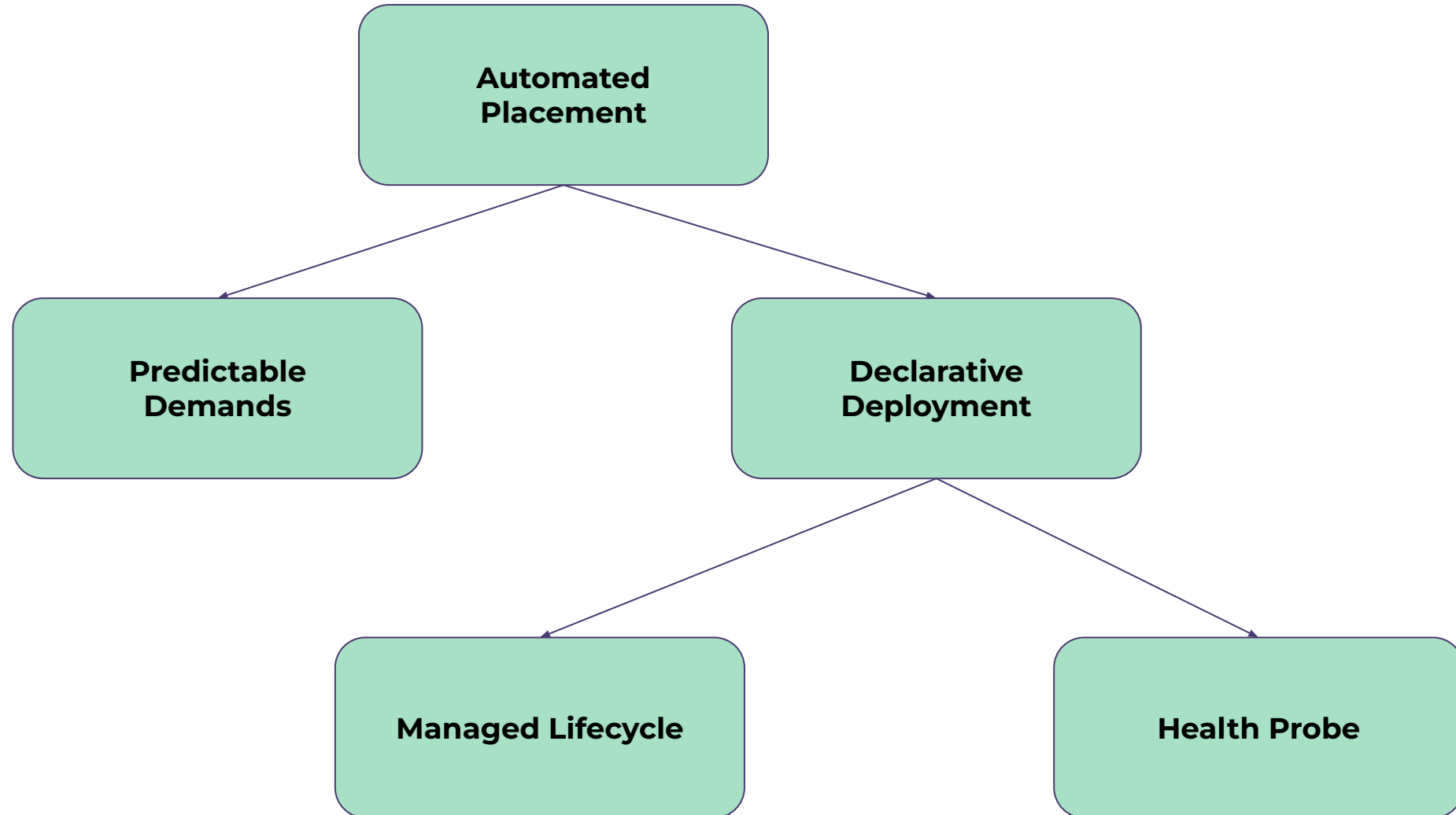
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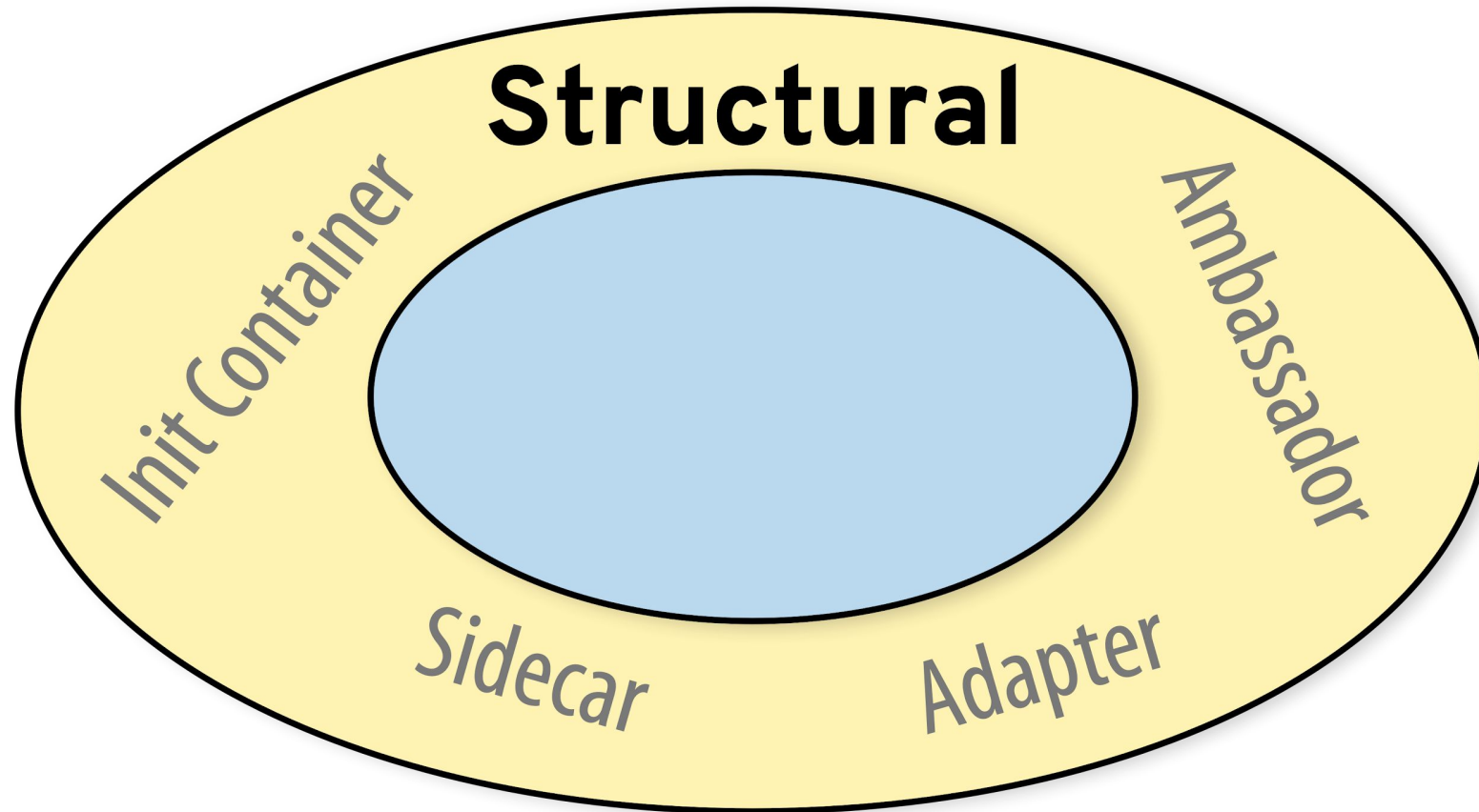


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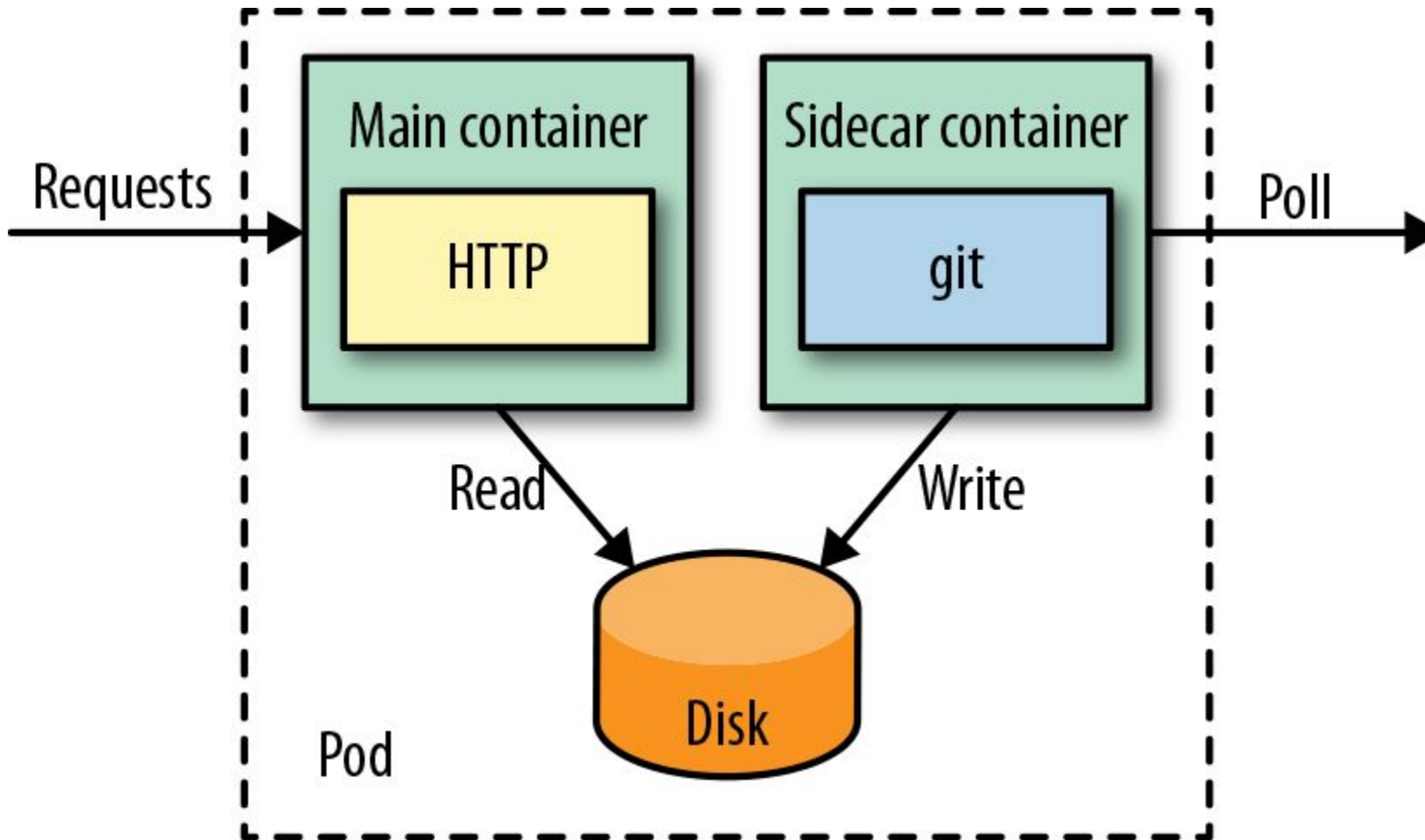


A photograph of a sidecar motorcycle race on a muddy track. In the foreground, a rider in a maroon and white suit and helmet is driving a sidecar. A passenger in a black suit and helmet is seated in the sidecar, holding onto the driver. The sidecar is a wooden box-like structure. In the background, two other riders on motorcycles are visible, also on the muddy track. The track is surrounded by green grass and a white fence. The word "Sidecar" is overlaid in large white text on a dark horizontal band across the middle of the image.

Sidecar

How to enhance the
functionality of an
application without
changing it

- Runtime collaboration of containers
- Connected via shared resources:
 - Network
 - Volumes
- Similar what AOP is for programming
- Separation of concerns



Startup Order



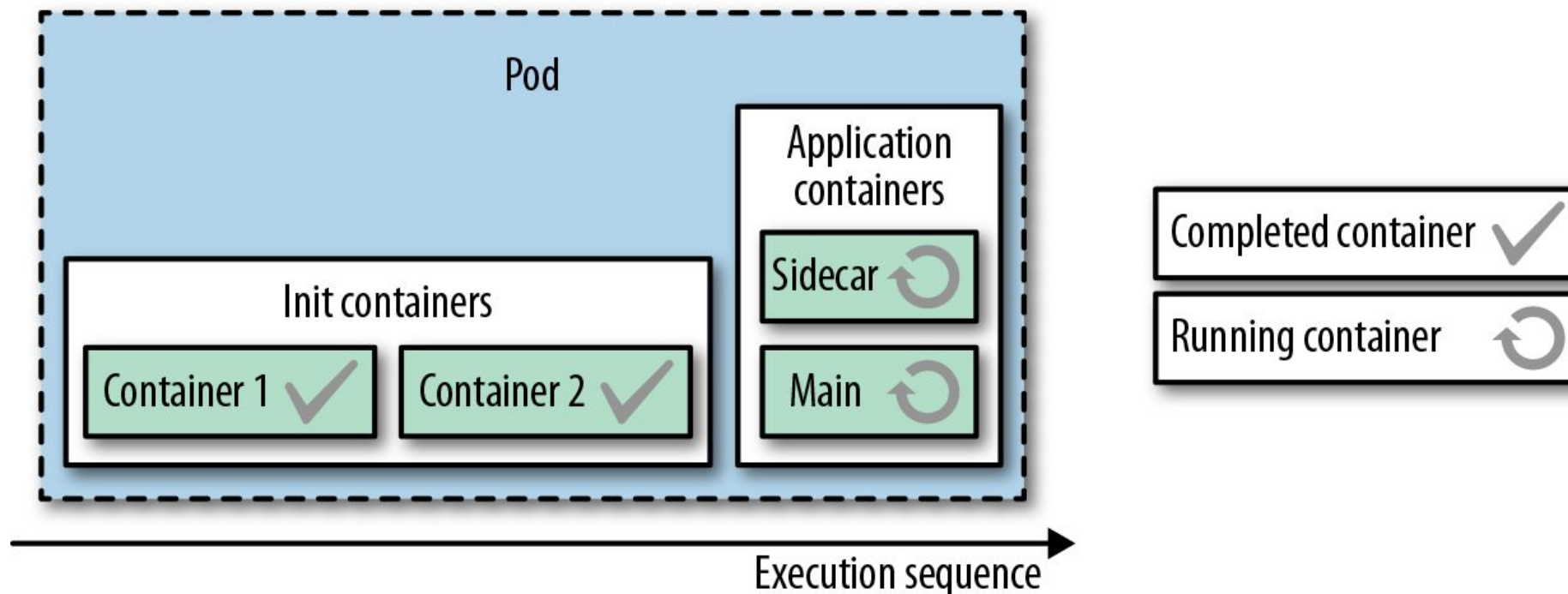
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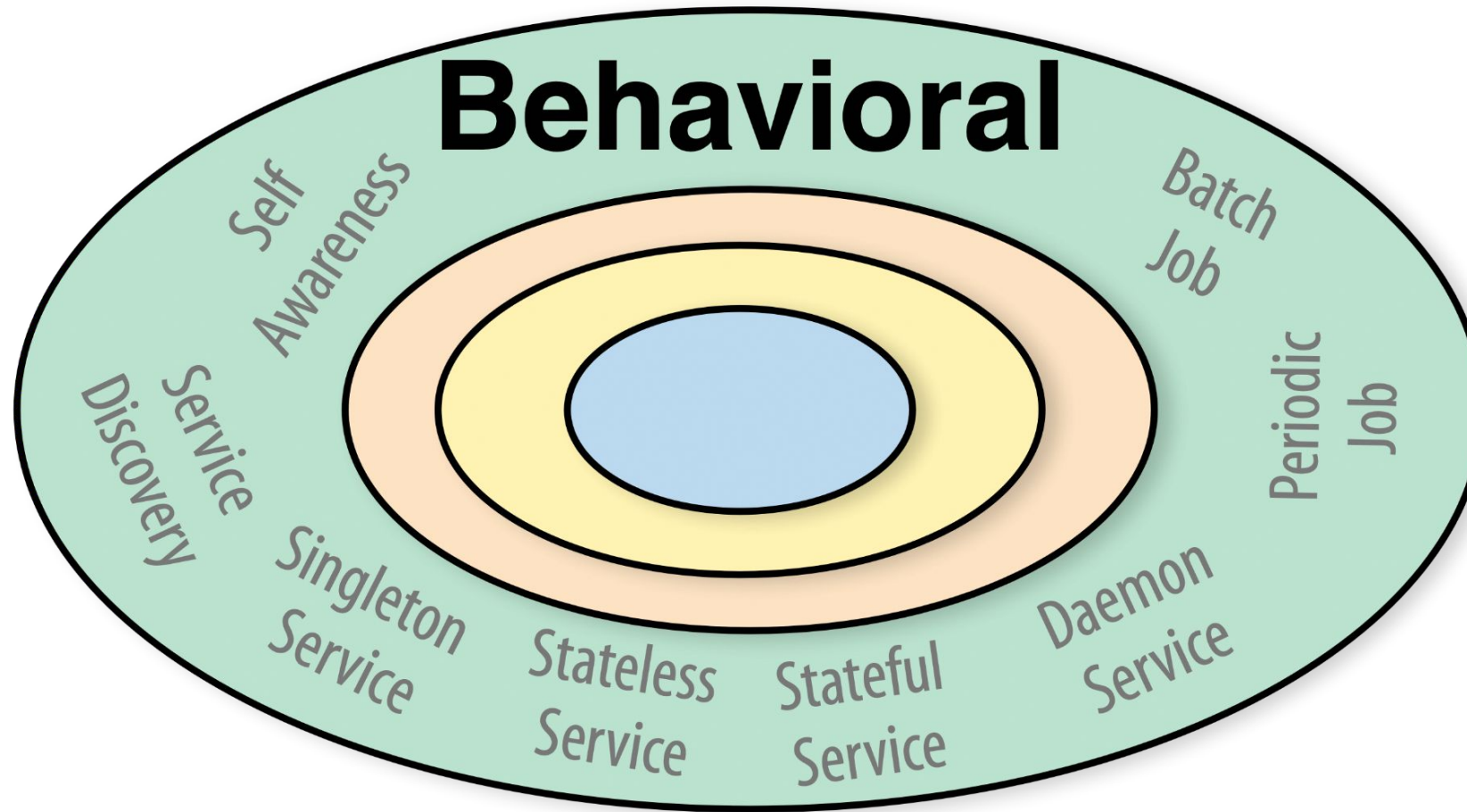
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- Dedicated sidecar containers (since 1.29, beta):
 - `initContainers[].restartPolicy == Always`
 - Continues to run after init containers have finished
 - Supports probes

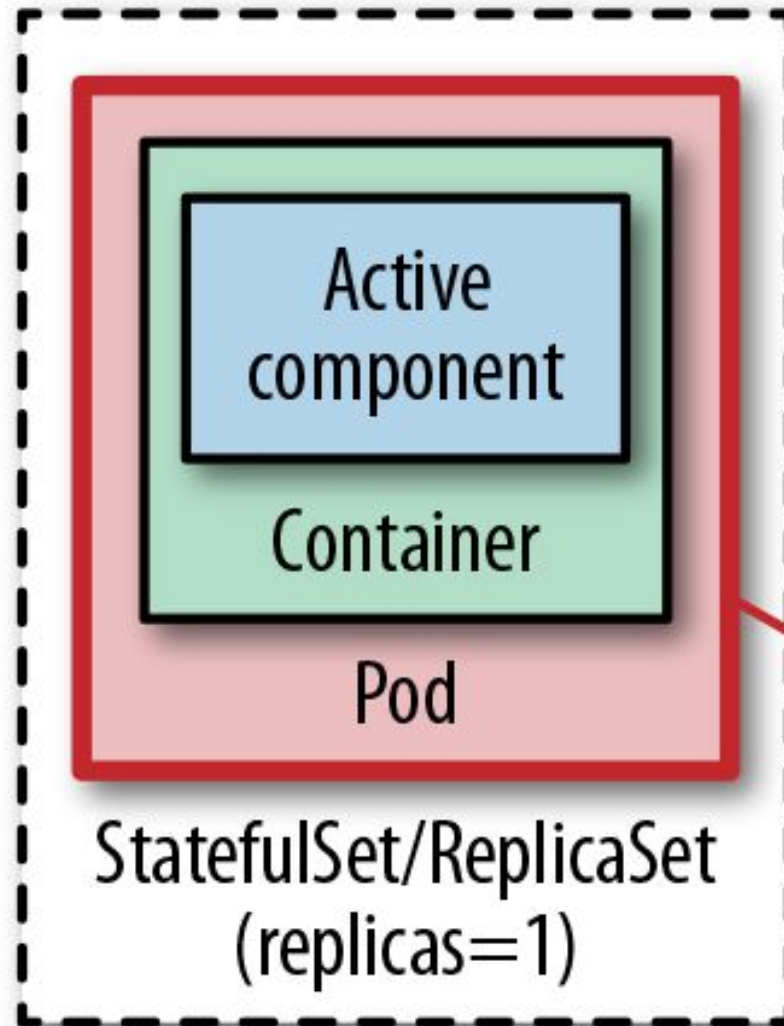


Singleton Service



**How to ensure that
only one application
instance is active**

Out-of-Application Locking



```
kind: ReplicaSet
metadata:
  name: file-poller
spec:
  serviceName: file-poller
  replicas: 1
```

Singleton

- ReplicaSet with **(at least)** 1 replica
 - Highly available Pod which is monitored and restarted in case of failures
 - Favors availability over consistency
 - more than one Pod can exist temporarily
- StatefulSet with **(at most)** 1 replica
 - Favors consistency over availability
 - Favors availability over consistency
 - less than one Pod can exist temporarily

In-Application Locking



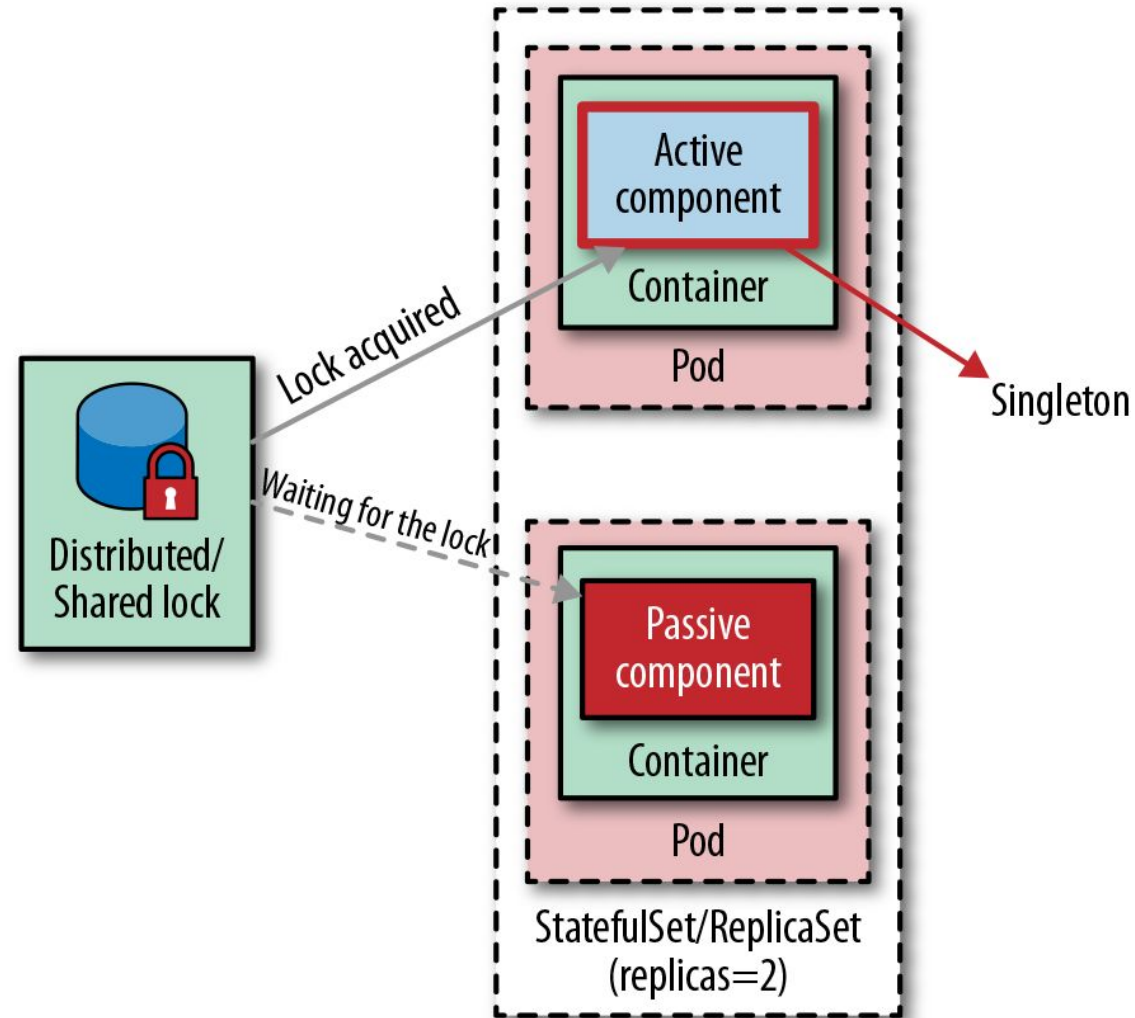
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- Distributed lock shared by simultaneously running applications
- *Active-Passive* topology
- Distributed lock implementations e.g.
 - Zookeeper
 - Consul
 - Redis
 - etcd,
 - Kubernetes ConfigMaps, Lease resource

Distributed lock API



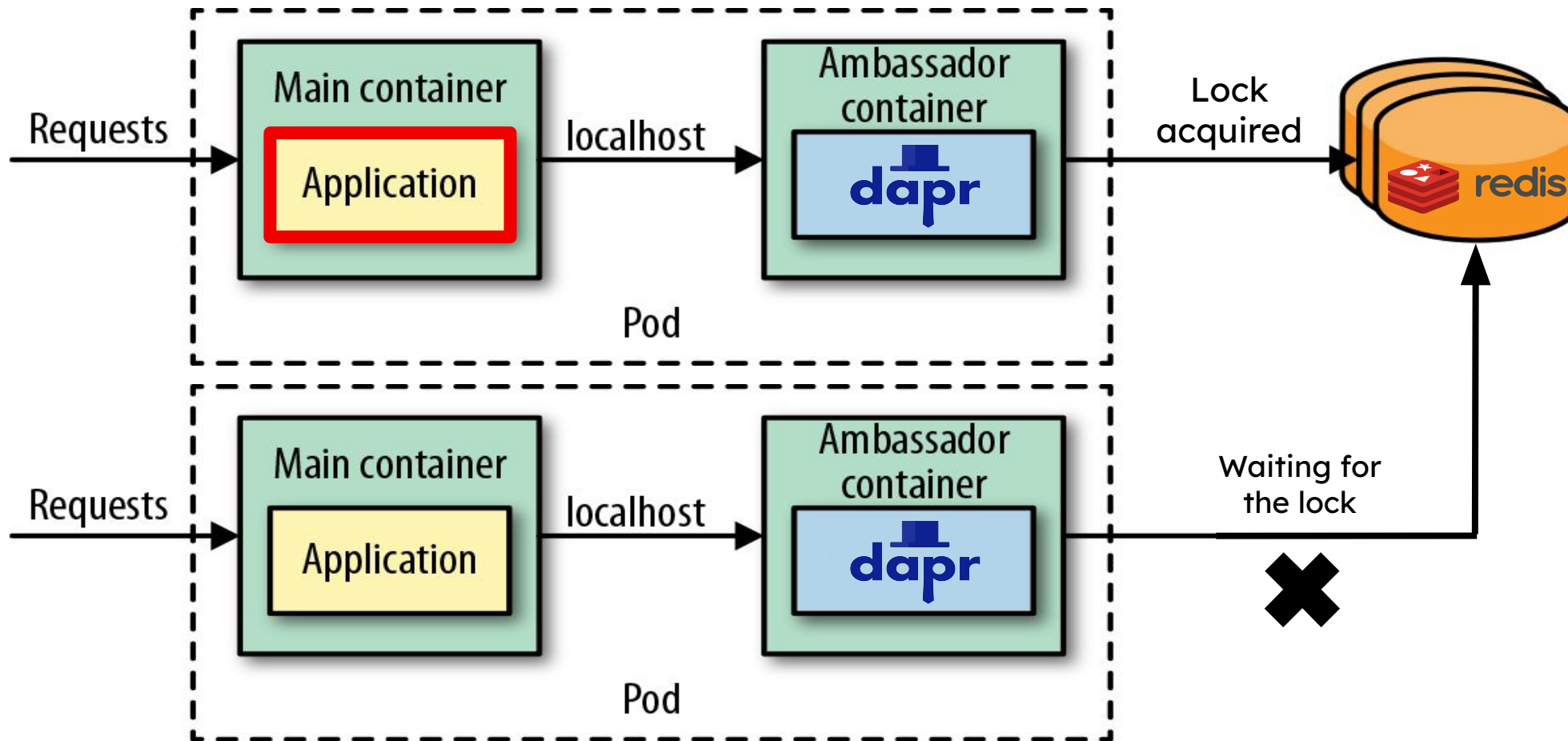
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```
curl -X POST http://localhost:3500/v1.0-alpha1/lock/lockstore -H 'Content-Type: application/json' -d '{"resourceId":"my_file_name", "lockOwner":"abc123", "expiryInSeconds": 60}'
```

Configuration Patterns



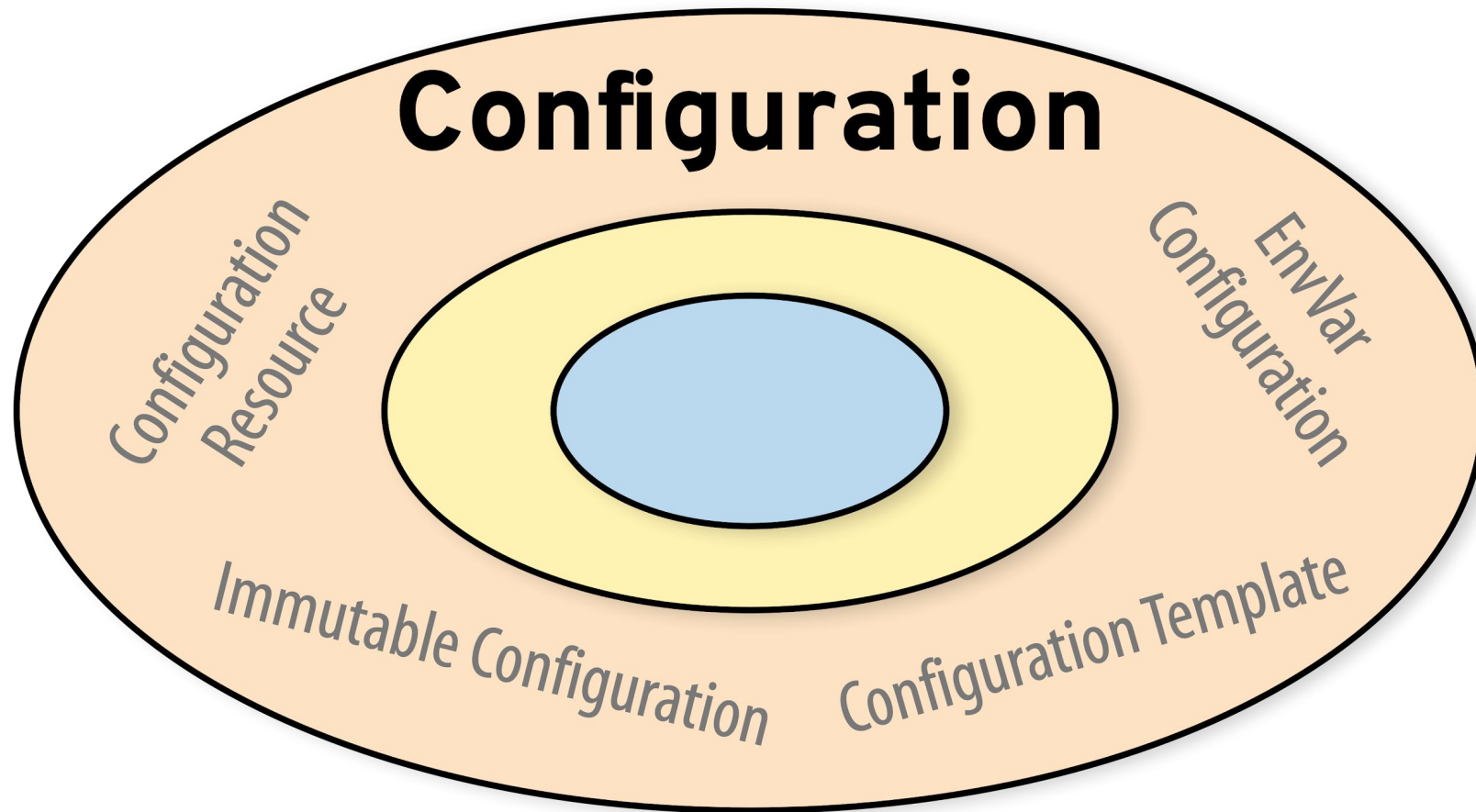
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A large, rounded rock formation, known as Sea Stack, stands prominently in the ocean. The rock is a reddish-brown color with visible geological textures and some sparse vegetation. The sky above is a clear, bright blue. The water in the foreground is a deep blue with gentle ripples. The text "Immutable Configuration" is overlaid in a large, white, sans-serif font, centered horizontally and partially covering the rock formation.

Immutable Configuration

How to configure your application with immutable container images

- Configuration that can **not be changed**
- Different sets of configuration for **different environments** (dev/prod)
- **Versioning** and **auditing** for configuration
- **Solution:** Configuration stored in **container images**
 - Versioned via tags and digests
 - Distributed via image registry

Configuration provided by Init Container



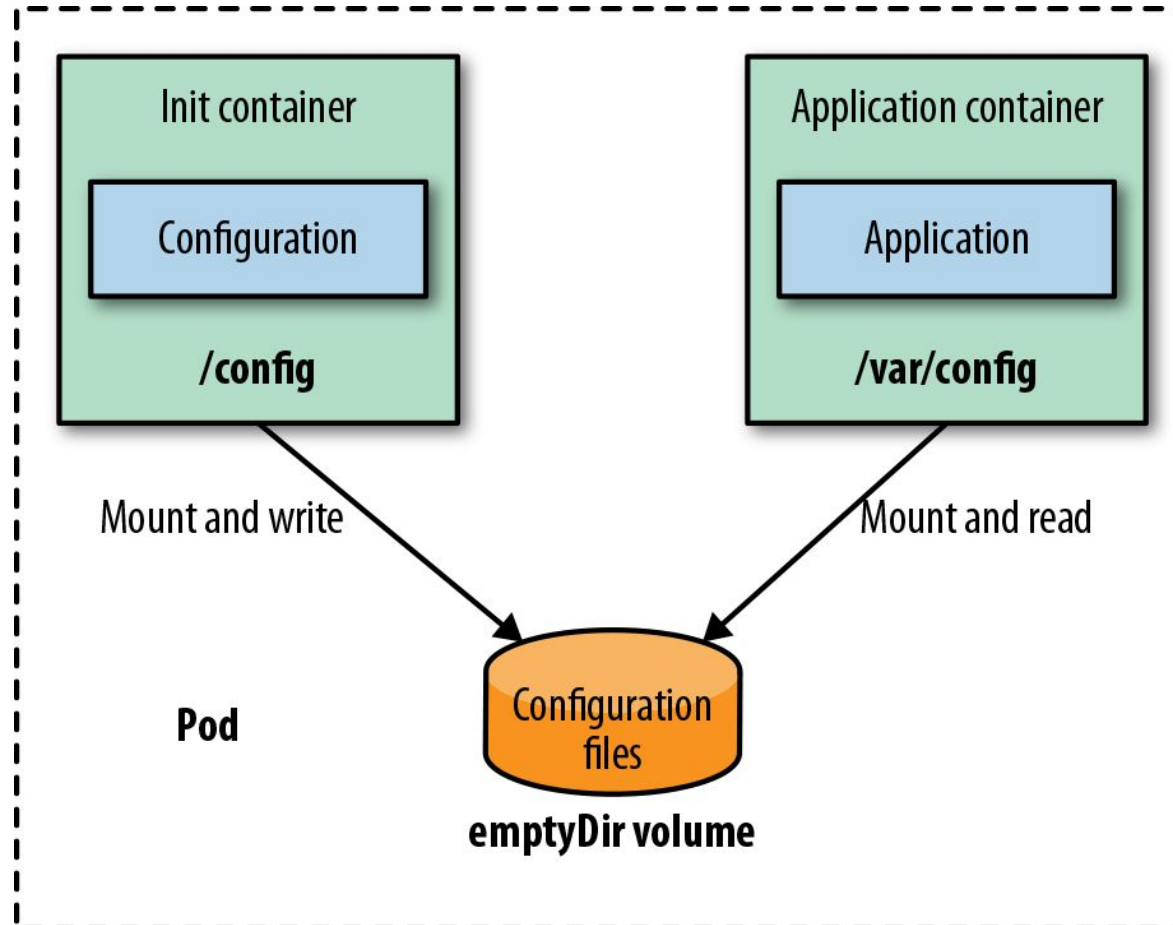
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Directly sharing the container filesystem



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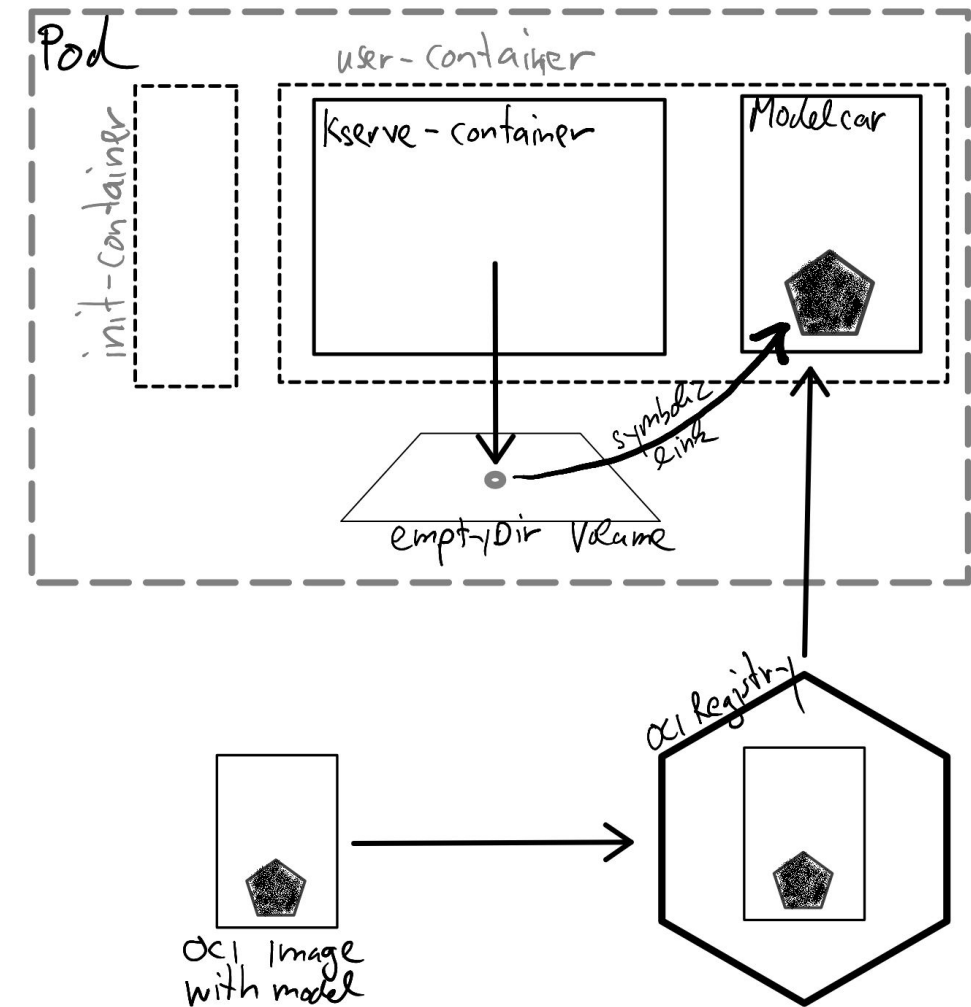


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- Sharing the process namespace, including the file system via **shareProcessNamespace: true**
- Technique for directly accessing another containers filesystem via **/proc/<pid>/root**
- Used in *modelcars* in KServe for accessing large language models



Security Patterns



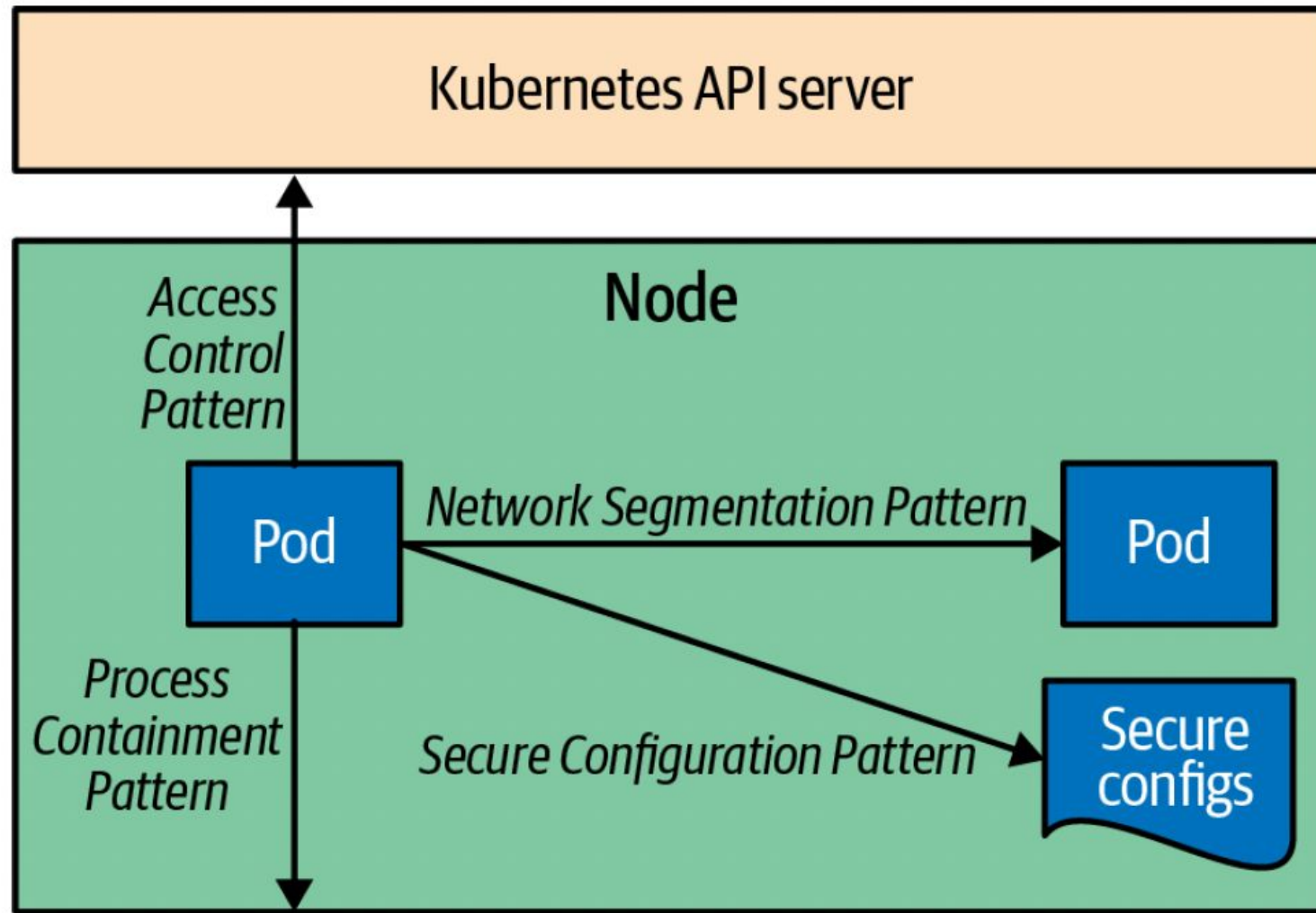
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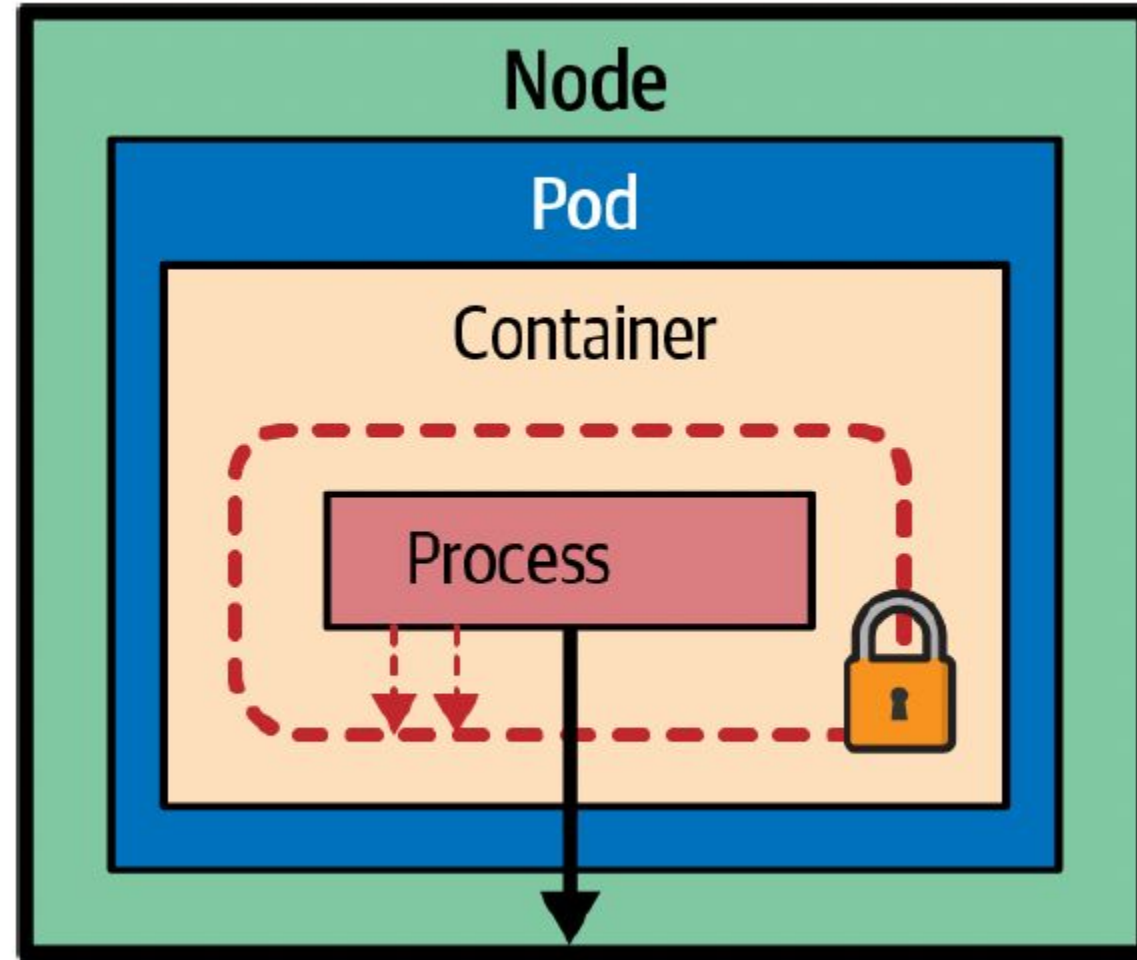
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Process Containment

How to protect the platform against deployed code



Run Containers with a Non-Root User



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Forces any container in the Pod to run with user ID 1000 and group ID 2000

```
apiVersion: v1
kind: Pod
metadata:
  name: web-app
spec:
  securityContext:
    runAsUser: 1000
    runAsGroup: 2000
  containers:
  - name: app
```

Prevent any container from starting with a root user—that is, a user with UID 0.

```
apiVersion: v1
kind: Pod
metadata:
  name: web-app
spec:
  securityContext:
    runAsNonRoot: true
  containers:
  - name: app
```

Lock the Capabilities



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Prevents privilege escalation

```
apiVersion: v1
kind: Pod
metadata:
  name: web-server
spec:
  containers:
  - name: httpd
    image: httpd
    securityContext:
      allowPrivilegeEscalation: false
```

Reduce capabilities to required ones only

```
apiVersion: v1
kind: Pod
metadata:
  name: web-server
spec:
  securityContext:
    capabilities
      drop: ['ALL']
      add: ['NET_BIND_SERVICE']
  containers:
  - name: httpd
```

Avoiding a Mutable Container Filesystem



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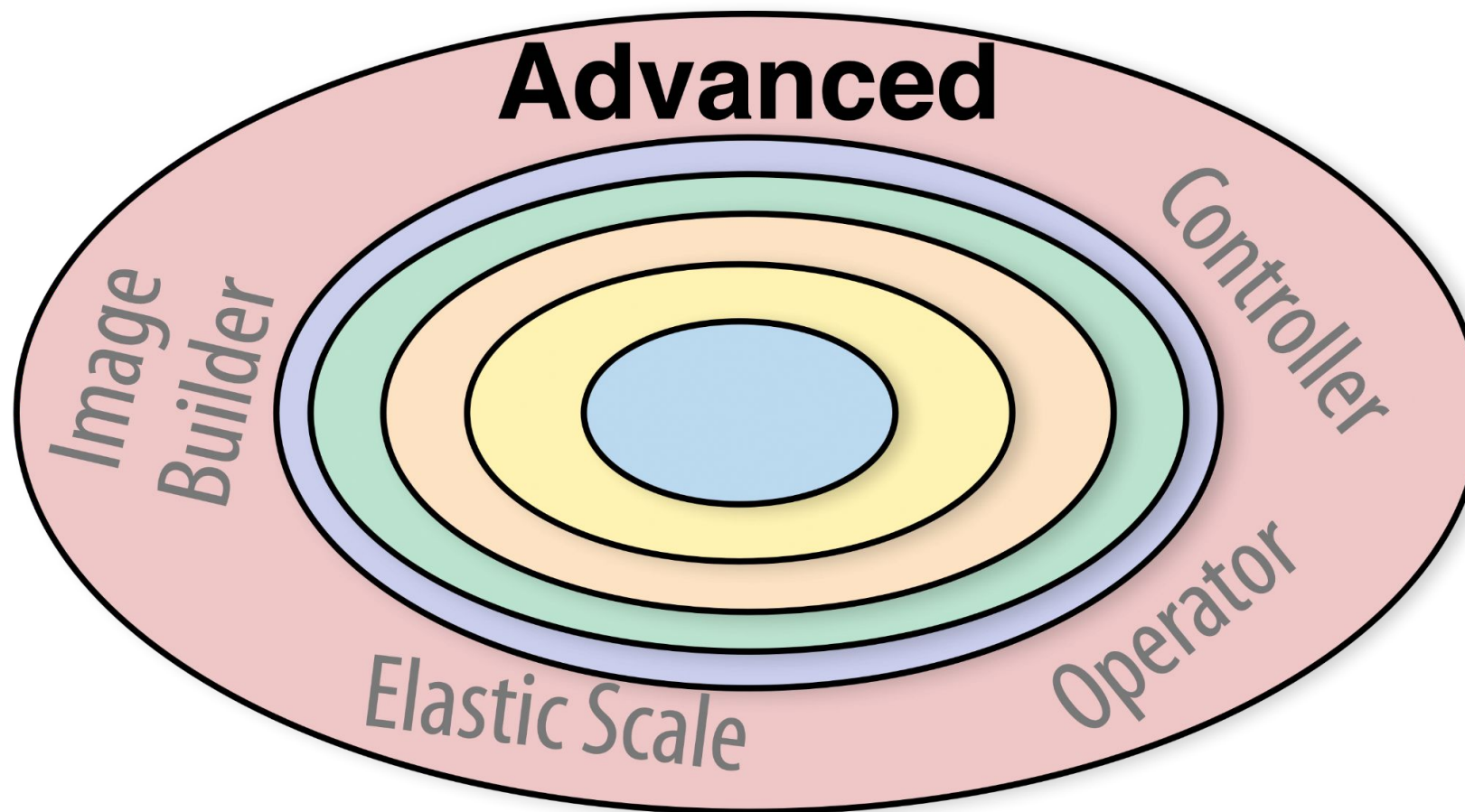
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Set read only file system

```
apiVersion: v1
kind: Pod
metadata:
  name: web-server
spec:
  containers:
  - name: httpd
    image: httpd
    securityContext:
      readOnlyRootFilesystem: true
```

Overview of options we covered:

```
apiVersion: v1
kind: Pod
metadata:
  name: web-server
spec:
  containers:
  - name: httpd
    image: httpd
    securityContext:
      runAsNonRoot: true
      allowPrivilegeEscalation: false
      readOnlyRootFilesystem: true
      capabilities:
        drop: - ALL
        add: ['NET_BIND_SERVICE']
```

Controller



**How to get from the
current state to a
declared target state**

- Kubernetes as distributed state manager
- Make the **actual** state more like the declared **target** state.



Observe - Discover the actual state



Analyze - Determine difference to target state



Act- Perform actions to drive the actual to the desired state

Observe - Analyze - Act



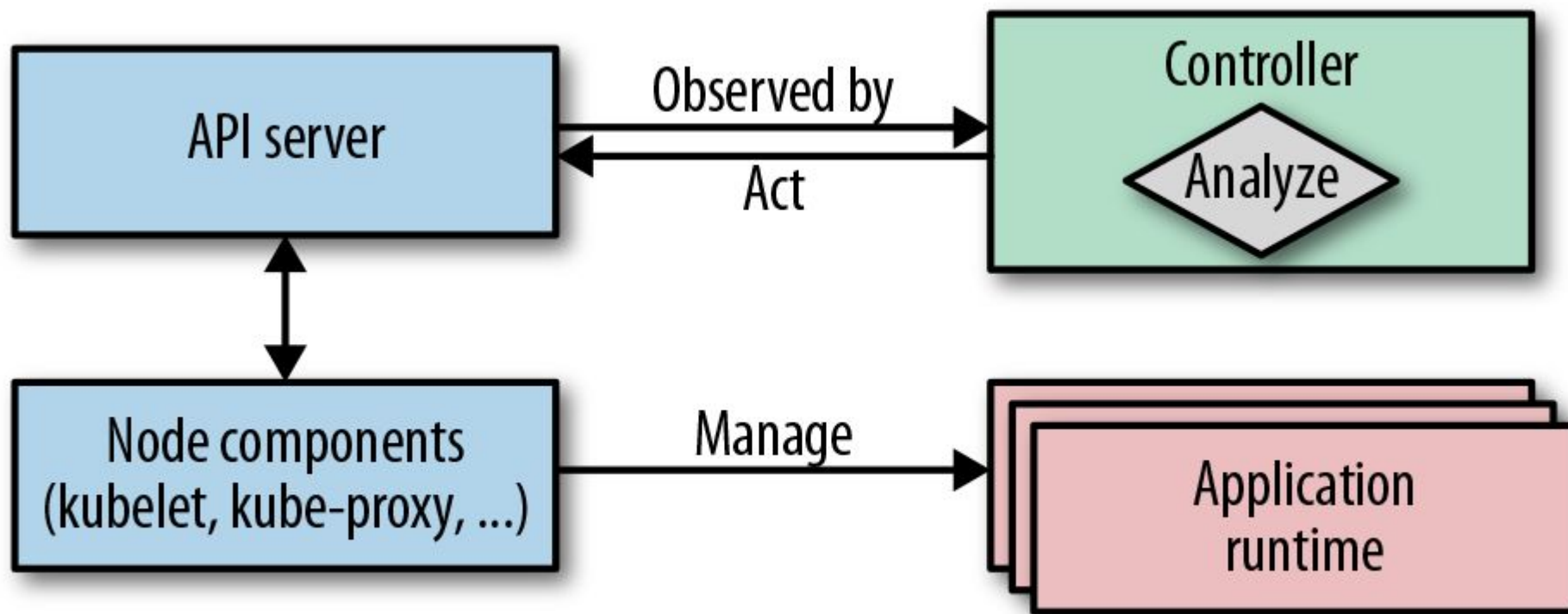
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Operator

How to encapsulate operational knowledge into executable software

“““ An **operator** is a Kubernetes **controller** that understands two domains: Kubernetes and *something else*. By combining knowledge of both areas, it can **automate** tasks that usually require a human operator that understands both domains.

Jimmy Zelinskie

<http://bit.ly/2Fjlx1h>

Operator = Controller + CustomResourceDefinition

Custom resource is modelling a custom domain and managed through the Kubernetes API

```
apiVersion: apiextensions.k8s.io/v1beta1
kind: CustomResourceDefinition
metadata:
  name: configwatchers.k8spatterns.io
spec:
  scope: Namespaced
  group: k8spatterns.io
  version: v1
  names:
    kind: ConfigWatcher
    plural: configwatchers
  validation:
    openAPIV3Schema:
      ...
```

```
kind: ConfigWatcher
apiVersion: k8spatterns.io/v1
metadata:
  name: webapp-config-watcher
spec:
  configMap: webapp-config
  podSelector:
    app: webapp
```

Operator



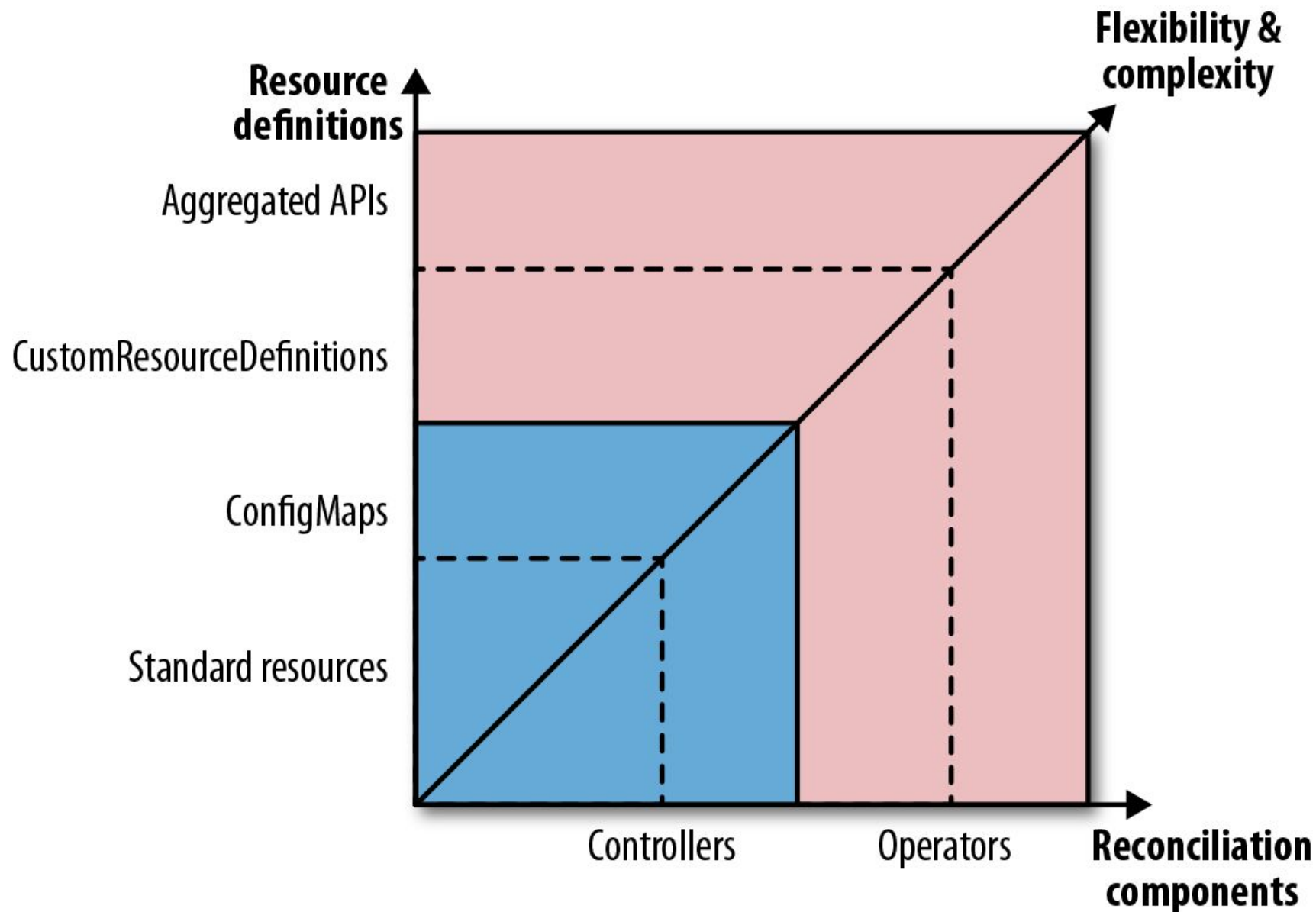
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- Installation CRDs
 - Installing and operating applications
 - Backup and Restore
 - Monitoring and self-healing
 - Example: Prometheus for installing Prometheus & components
- Application CRDs
 - Application specific domain concepts
 - Example: ServiceMonitor for registering Kubernetes service to be scraped by Prometheus

OperatorHub.io | The registry

OperatorHub.io

Search OperatorHub...

Contribute

Welcome to OperatorHub.io

OperatorHub.io is a new home for the Kubernetes community to share Operators. Find an existing Operator or list your own today.

PROVIDER

☐ Amazon Web Services (1)

☐ CNCF (1)

☐ Couchbase (1)

☐ Crunchy Data Solutions (1)

☐ Dynatrace (1)

☒ Jaeger (1)

☒ MongoDB (1)

☐ Percona (1)

☐ PlanetScale (1)

☒ Red Hat (3)

☐ Redis Labs (1)

Show less

CAPABILITY LEVEL

☐ Basic Install (4)

☐ Seamless Upgrades (3)

☐ Full Lifecycle (6)

5 ITEMS

VIEW

SORT A-Z



Jaeger Tracing

provided by Jaeger

Provides tracing, monitoring and troubleshooting microservices-based



Kubernetes Federation

provided by Red Hat

Gain Hybrid Cloud capabilities between your clusters with Kubernetes Federation.



MongoDB

provided by MongoDB, Inc

The MongoDB Enterprise Kubernetes Operator enables easy deploys of MongoDB



Prometheus Operator

provided by Red Hat

The Prometheus Operator for Kubernetes provides easy monitoring definitions for

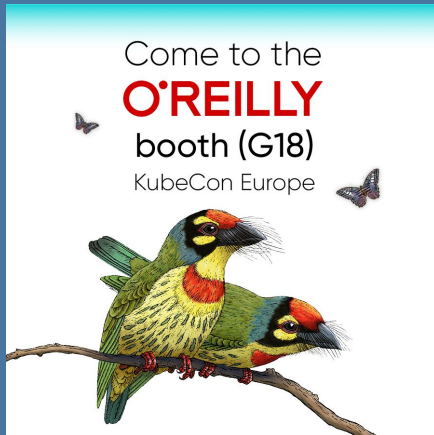


Strimzi Kafka

provided by Red Hat

Run an Apache Kafka cluster, including Kafka Connect, ZooKeeper and more.

May the duck be with you ...



Wednesday, 13:00



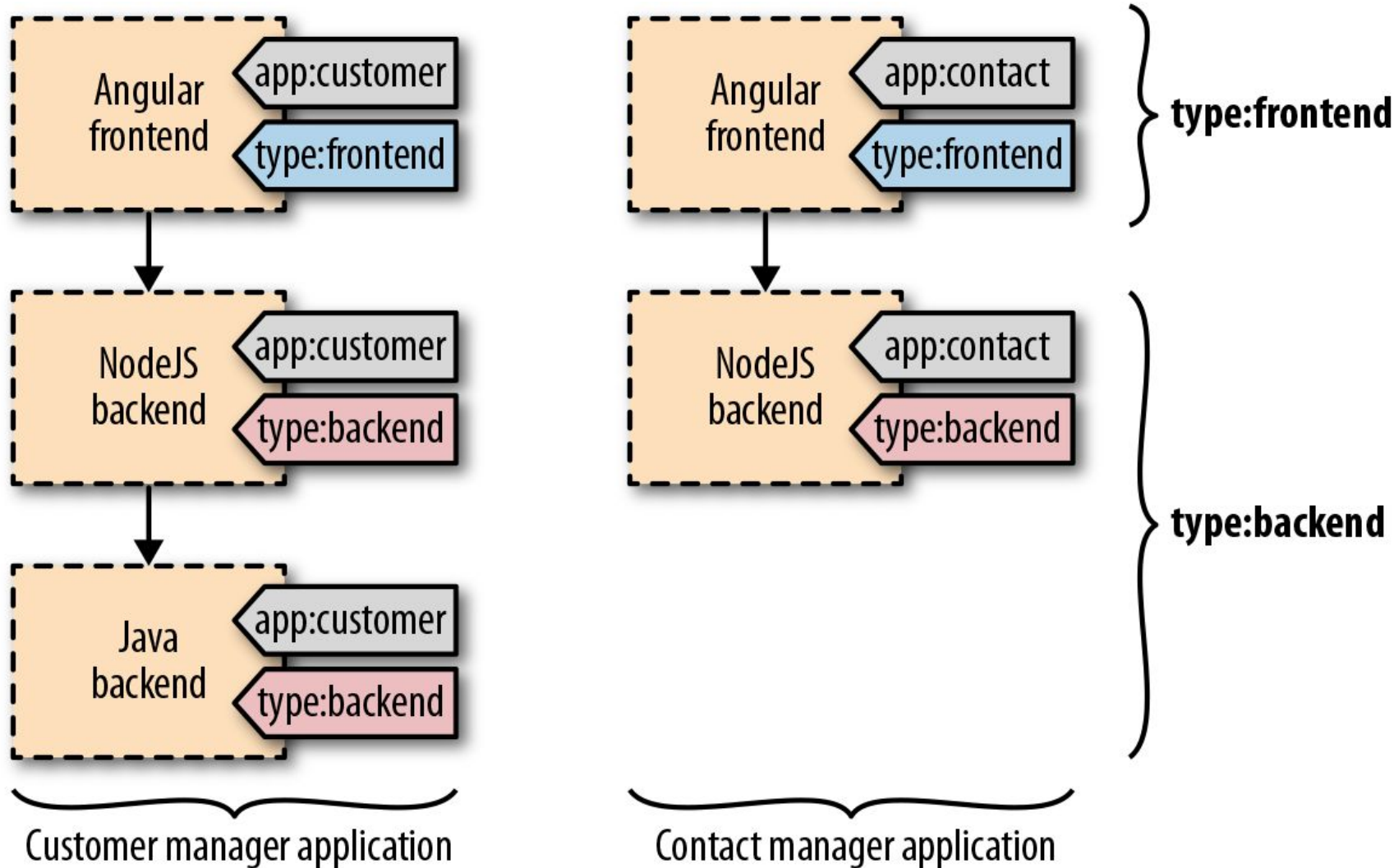
Thursday, 12:30

<https://k8spatterns.com>

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<https://pixabay.com/images/id-407101/>
<https://www.pexels.com/photo/backlit-beach-dawn-dusk-588561/>



Local vs. Distributed Primitives

Concept	Local primitive	Distributed primitive
Behavior encapsulation	Class	Container image
Behavior instance	Object	Container
Unit of reuse	<i>.jar</i>	Container image
Composition	Class A contains Class B	Sidecar pattern
Inheritance	Class A extends Class B	A container's FROM parent image
Deployment unit	<i>.jar/.war/.ear</i>	Pod
Buildtime/Runtime isolation	Module, package, class	Namespace, Pod, container
Initialization preconditions	Constructor	Init container
Postinitialization trigger	Init-method	<code>postStart</code>
Predestroy trigger	Destroy-method	<code>preStop</code>
Cleanup procedure	<code>finalize()</code> , shutdown hook	-
Asynchronous and parallel execution	<code>ThreadPoolExecutor</code> , <code>ForkJoinPool</code>	Job
Periodic task	<code>Timer</code> , <code>ScheduledExecutorService</code>	CronJob
Background task	Daemon thread	DaemonSet
Configuration management	<code>System.getenv()</code> , <code>Properties</code>	ConfigMap, Secret

Which Implementation to Choose?



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- Out-of-Application Locking
 - If you have good control at runtime
- In-Application Locking
 - If you need advanced features and customization
- Distributed lock API
 - Universal lock API for any language

Docker Volume Mount



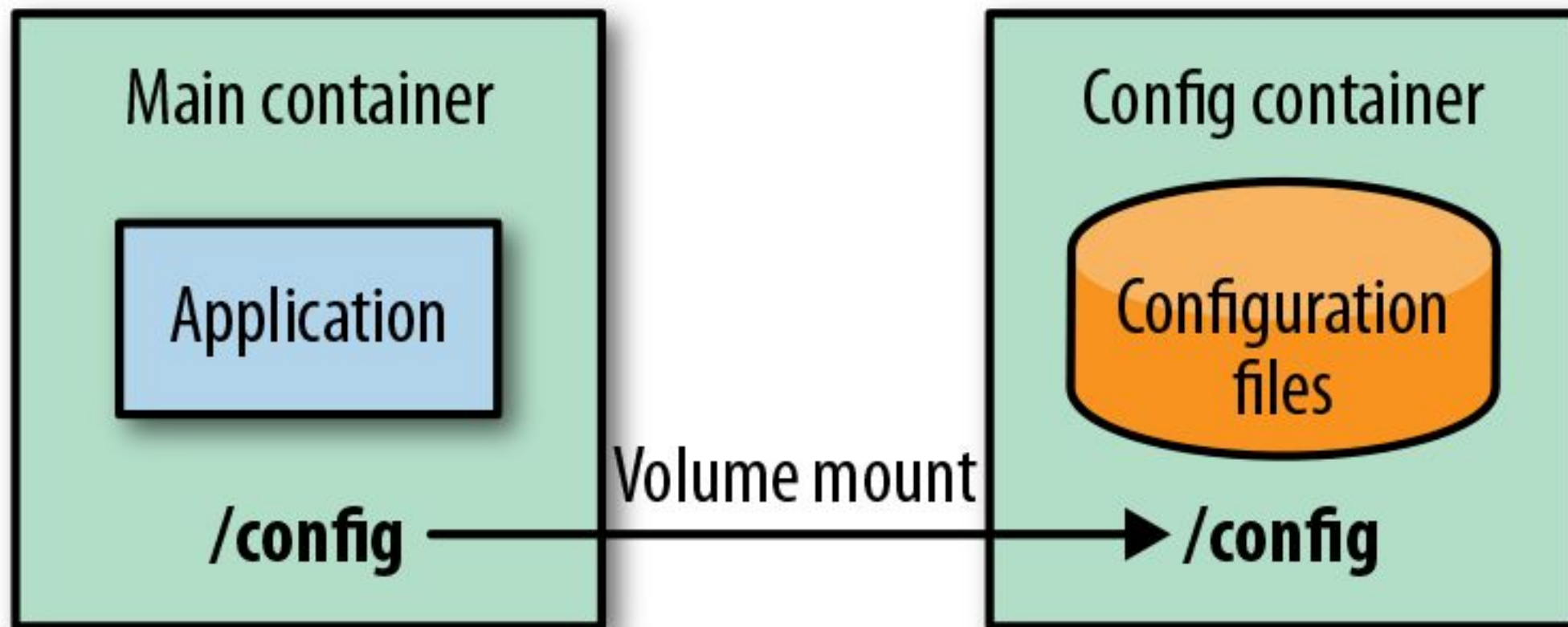
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Example



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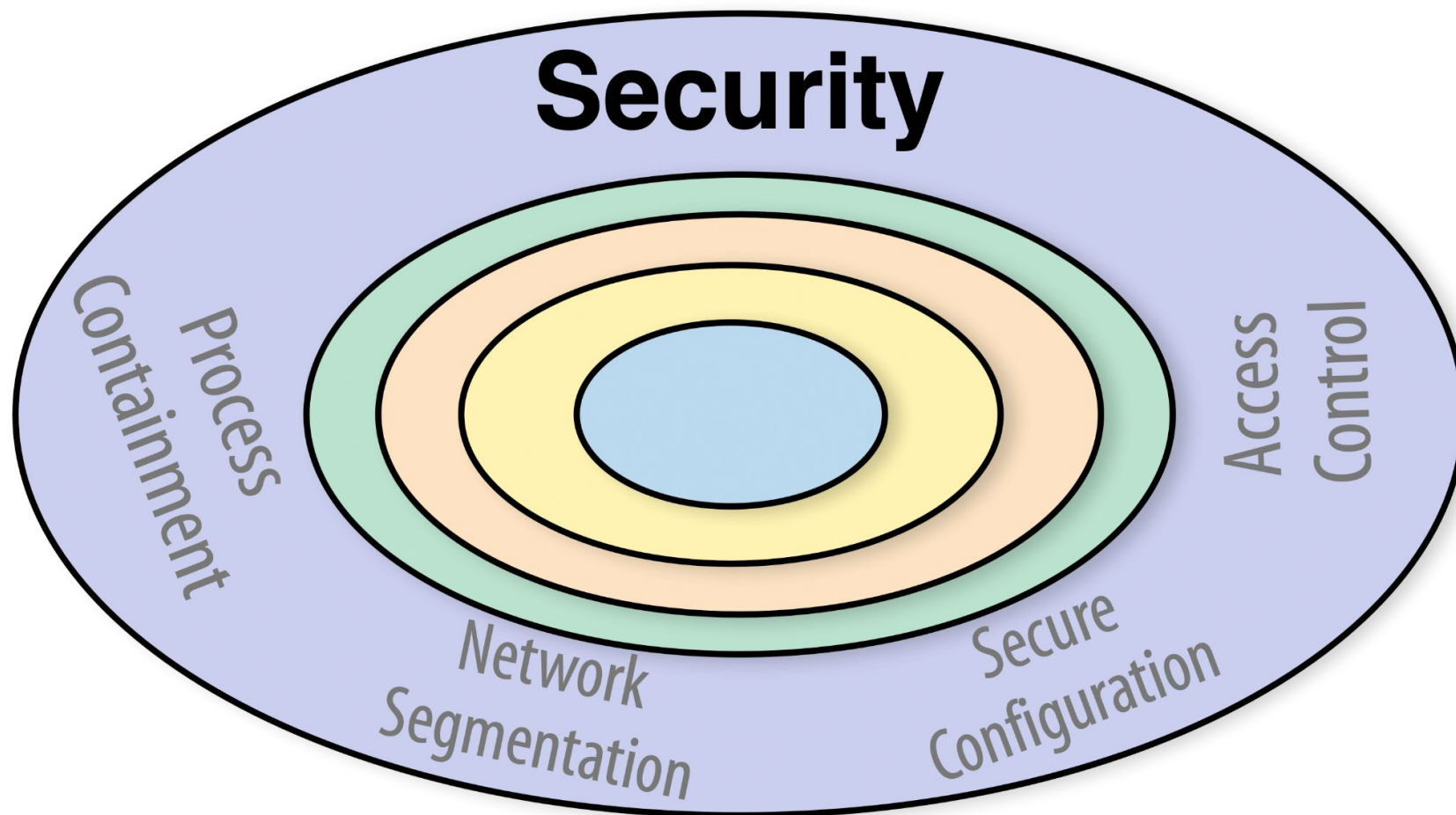
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```
initContainers:
- image: k8spatterns/config-dev:1
  name: init
  args:
  - "/config"
  volumeMounts:
  - mountPath: "/config"
    name: config-directory
containers:
- image: k8spatterns/demo:1
  name: demo
  ports:
  - containerPort: 8080
name: http
  protocol: TCP
  volumeMounts:
  - mountPath: "/config"
    name: config-directory
volumes:
- name: config-directory
  emptyDir: {}
```



```
FROM busybox
ADD dev.properties /config-src/demo.properties
ENTRYPOINT \
  [ "sh", "-c", "cp /config-src/* $1", "--" ]
```

- Kubernetes: **No container image volume mounts**
- Instead: **Copy over configuration** from image to a shared volume in an ***Init Container***
- **Templating** of deployments for adapting to different environments
 - OpenShift Templates, Helm Charts, Operator with CRD ...
- Good for **large configuration data**

- Labels
 - Indexed by backend
 - Suitable for selector-like functionality
 - Limitation on character set for names and values
- Annotations
 - No syntax restrictions
 - Not indexed
- ConfigMaps
 - Good for complex structured state declarations
 - Simple alternative to CustomResourceDefinitions

- Operator can be implemented in any language
- Frameworks:
 - Operator Framework (Golang, Helm, Ansible, Java)
<https://github.com/operator-framework>
 - Kubebuilder (Golang)
<https://github.com/kubernetes-sigs/kubebuilder>
 - Metacontroller (Language agnostic)
<https://metacontroller.app/>

ConfigMap Watch Controller



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```
namespace=${WATCH_NAMESPACE:-default}
base=http://localhost:8001
ns=namespaces/$namespace

curl -N -s $base/api/v1/${ns}/configmaps?watch=true | \
while read -r event
do
    type=$(echo "$event" | jq -r '.type')
    config_map=$(echo "$event" | jq -r '.object.metadata.name')
    annotations=$(echo "$event" | jq -r '.object.metadata.annotations')

    if [ $type = "MODIFIED" ]; then
        # Restart Pods using this ConfigMap
        # ...
    fi
done
```

- 4 R Intro (Patterns, Kubernetes) - 5 min
- 5 B HealthProbe
- 4 R Sidecar
- 5 B Singleton
- 5 R Immutable Configuration
- 4 B Process Containment
- 7 R Controller / Operator
- 1 B Summary